AI in Civil Engineering

ORIGINAL ARTICLE

Open Access

Optimized machine learning algorithms with SHAP analysis for predicting compressive strength in high-performance concrete

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Abstract

This research examines the application of eight different machine learning (ML) algorithms for predicting the compressive strength of high-performance concrete (HPC). Achieving precise predictions is crucial for enhancing structural reliability and optimizing resource usage in construction projects. The analysis utilized the "Concrete Compressive Strength" dataset, sourced from UC Irvine's publicly available ML repository. The models evaluated include Gradient Boosting Regressor (GBR), Extreme Gradient Boosting Regression (XGBoost), Random Forest (RF), Support Vector Regression (SVR), Artificial Neural Network (ANN), Multilayer Perceptron (MLP), Lasso, and k-Nearest Neighbors (KNN). To enhance performance, critical data preprocessing steps were undertaken, which involved feature scaling, cleaning, and normalization. Hyperparameter tuning via Grid Search (GS) and K-fold cross-validation further optimized the models. Among those analyzed, XGBoost and GBR achieved the highest predictive accuracy, with R^2 values of 93.49% and 92.09% respectively, coupled with lower mean squared error (MSE), mean absolute error (MAE), and root mean squared error (RMSE). SHapley Additive exPlanations (SHAP) analysis revealed cement content and curing age as the most significant factors affecting compressive strength. Validation against experimental data confirmed the reliability of XGBoost and GBR through consistent prediction patterns and close alignment with empirical measurements. The results establish ML as an effective approach for HPC strength prediction, offering advantages in computational efficiency and accuracy over conventional analytical methods.

Keywords Concrete strength, Machine learning, Prediction, Boosting, Regression, High performance concrete, Hyperparameter tuning, Grid search, Cross validation, SHapley Additive exPlanations

1 Introduction

Conventional concrete, composed of fine/coarse aggregates bonded by cementitious materials, has served as a fundamental construction material for decades. While cement remains essential in modern formulations, contemporary concrete constitutes a composite system integrating mineral additives and chemical admixtures to achieve enhanced properties beyond traditional mixtures (Aïtcin, 2000).

High-performance concrete (HPC) distinguishes itself through supplementary cementitious materials (fly ash, silica fume, blast furnace slag) and superplasticizers, which collectively improve mechanical strength while reducing water-to-binder ratios (Muhammad et al., 2024). These characteristics make HPC particularly suitable for demanding applications including skyscrapers, defense installations, prefabricated structures, and bridge engineering. However, the complex heterogeneous composition of HPC presents formulation challenges (Lee et al., 2022).

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Concrete is widely utilized in engineering structures globally, and its technology is continuously evolving through innovations and advancements. Predicting concrete's strength is immensely advantageous. Accurate strength prediction not only ensures the safety and permanence of HPC structures but also aids in optimizing material usage and improving construction efficiency. Traditional methods for determining concrete strength often depend on empirical formulas and laboratory tests, which can be labor-intensive, costly, and sometimes imprecise. Standard testing approaches using cube or cylindrical samples may exhibit variability due to differences between laboratory conditions and real-world applications, as well as time-consuming procedures. Non-destructive testing methods like the Schmidt hammer or pulse velocity tests offer quicker and more cost-effective alternatives but may not always deliver precise estimates of internal concrete strength.

This study explores the innovative potential of Machine Learning (ML) in forecasting concrete strength. As a subfield of artificial intelligence (AI), ML employs data-driven methodologies to uncover patterns, trends, and correlations within vast datasets, enabling precise predictions. By leveraging advanced ML algorithms for prediction, engineers can utilize data-driven insights to accurately forecast compressive strength. These algorithms offer a dynamic approach by scrutinizing extensive datasets, capturing intricate interactions between parameters that traditional methods might overlook. ML algorithms excel at recognizing nonlinear relationships and patterns within data, thereby bolstering predictive capabilities. The Supplementary Values plots (SHAP) analysis further enhances this process, providing a fresh perspective on examining feature importance and impact, granting deeper insights into the influence of individual parameters on predictions (Huang et al., 2020). By analyzing the unique strengths of algorithms such as GBR, XGBoost, Lasso, KNN, and others, this research underscores the potential of ML in transforming HPC applications. Furthermore, the study explores how advanced techniques like hyperparameter tuning can enhance model performance, resulting in unprecedented accuracy in forecasting concrete strength. The integration of ML into contemporary concrete technologies heralds a new era of innovation, providing the construction sector with more efficient and reliable tools.

This study uniquely incorporates SHAP to evaluate individual feature effects and contributions in predicting the compressive strength of HPC, offering novel insights into feature contributions. Recent studies have demonstrated that SHAP can be employed to determine the impact of individual features and their significance, whether positive or negative. By quantifying the influence of each feature, SHAP offers a clearer comprehension of how input variables such as cement content, age, and water-cement ratio affect the dependent output. SHAP introduces an innovative approach to optimizing concrete mix proportions, reducing the number of trials required to attain the desired strength, and ultimately conserving resources. This approach leads to resource conservation, enhances the efficiency and sustainability of concrete mix design practices, and ensures a more environmentally friendly construction process (Islam et al., 2024a).

1.1 Novelty and Contributions

In recent years, there has been a substantial progression in the application of machine learning (ML) to predict the compressive strength of high-performance concrete (HPC). A multitude of scholarly works have concentrated on the evaluation of diverse algorithms and the enhancement of their predictive accuracy through optimization strategies, including hyperparameter tuning and cross-validation. Despite these advancements, there remains an imperative need for a systematic exploration and comparative analysis of a broader spectrum of algorithmic approaches, particularly those adept at addressing the intricacies inherent in HPC datasets, such as the influence of supplementary materials that contribute to enhanced strength. Additionally, it is essential to scrutinize the impact of individual input features on the model output and to rigorously examine potential biases. This study addresses this research gap by evaluating eight carefully selected algorithms within a supervised learning paradigm, employing hyperparameter tuning and cross-validation techniques, utilizing the SHAP analysis to assess the influence of features, and analyzing their performance to elucidate detailed behavioral patterns. Through meticulous documentation and rigorous evaluation, this study provides valuable insights into the practical application of these models, thereby facilitating further advancements in the prediction of HPC compressive strength.

2 Overview of the ML algorithms and techniques

This study evaluates the efficacy of various machine learning algorithms by employing eight distinct models to predict the compressive strength of concrete. Each algorithm was selected for its capacity to handle regression tasks and to uncover complex data patterns. To bolster model robustness, hyperparameter tuning and cross-validation were employed to mitigate overfitting and to ensure the accuracy and reliability of predictions. Hyperparameters play a pivotal role in shaping ML models, influencing their architecture and the algorithms used to minimize loss functions. Examples include the regularization parameter C in support vector machines, the learning rate during neural network training, the choice of activation function, and the selection of optimizer (Yang & Shami, 2020). These strategies were crucial in improving the generalization ability of the models, thereby enhancing their accuracy in predicting compressive strength across diverse conditions.

2.1 Gradient boosting regressor

The Gradient Boosting Regressor (GBR), initially introduced by Friedman (2001), is a robust ensemble technique that constructs a sequence of decision trees, with each subsequent tree addressing the errors of its predecessor (Otchere et al., 2021). By employing gradient-based optimization methods, such as Stochastic Gradient Descent (SGD), GBR efficiently minimizes the loss function (Nguyen et al., 2021a, 2021b). It is capable of handling both continuous and categorical data, and its adaptability allows for the use of various loss functions, including the Huber loss, which mitigates the influence of outliers. Regularization techniques, such as shrinkage (learning rate) and subsampling, are instrumental in preventing overfitting.

Hyperparameter tuning is key to maximizing GBR's performance. Parameters like the number of estimators (n_estimators), depth of trees (max_depth), and learning rate are optimized through methods like RS and cross-validation. GBR is also effective in handling missing data and provides feature importance scores, which contribute to model interpretability. Its scalability makes it suitable for large datasets and is widely used in financial forecasting and reservoir characterization.

2.2 Extreme gradient boosting regressor

XGBoost, another highly efficient ensemble learning method, has gained widespread recognition for its exceptional predictive performance and computational speed (Chen et al., 2016). Particularly adept at handling sparse data, XGBoost is indispensable when confronted with diverse and intricate input variables, as commonly encountered in the realm of concrete strength prediction. Operating within a gradient boosting framework, XGBoost sequentially incorporates decision trees, each tree learning from the residual errors of its predecessor. This iterative process empowers the model to refine its predictions incrementally, culminating in superior accuracy. The algorithm's prowess in handling voluminous datasets replete with intricate patterns positions it as a formidable tool for high-performance predictive modeling applications across diverse disciplines (Chen & Guestrin, 2016a, 2016b).

Beyond its commendable performance, XGBoost has been fine-tuned to optimize speed and scalability. Its superior efficiency in processing extensive datasets can be attributed to its streamlined resource utilization, facilitating expedited computations without compromising precision. These enhancements encompass the integration of regularization techniques, like l_1 and l_2, which prevent overfitting and improve generalization on unseen data (Mesut et al., 2023). Noteworthy is XGBoost's adaptability in effectively managing missing data, while its seamless integration across various platforms further broadens its utility across diverse domains. The trifecta of adaptability, speed, and accuracy in navigating complex datasets underscores XGBoost's stature as a favored tool for multifaceted engineering applications *(*Chen & Guestrin, 2016a, 2016b*)*.

2.3 Random forest

Random Forest (RF), initially proposed by Breiman (2001), stands as a highly efficacious ensemble technique widely employed in machine learning to delineate intricate relationships and navigate extensive datasets with finesse. This approach engenders multiple decision trees by selecting random subsets of both data and features for each tree. The introduction of randomness mitigates the propensity for overfitting within individual trees, facilitating enhanced generalizability to novel data points. In the context of regression tasks, exemplified by concrete compressive strength prediction, Random Forest amalgamates the outputs from all trees, averaging them to derive a definitive prediction. Noteworthy is its resilience to missing data, along with its capacity to stave off both overfitting and underfitting, rendering it particularly well-suited to practical scenarios such as concrete strength estimation (Salman et al., 2024).

The algorithm is particularly valuable in capturing the intricate interactions between input variables, a crucial feature in predicting concrete strength. Factors such as material composition, environmental conditions, and curing time often interact in complex ways. RF has also demonstrated significant success in structural health (SHM), such as predicting the damage states of bridges during seismic events, with an accuracy rate exceeding 90%. Its efficacy in handling imbalanced datasets and addressing missing data gaps is exemplified in applications like monitoring the Forth Road Bridge (Xu et al., 2019). Furthermore, the algorithm's ability to learn from data without requiring substantial pre-processing or feature engineering enhances its practical usability in predictive tasks (Xu et al., 2023).

Although RF is highly accurate, it can be computationally intensive, particularly when training on large datasets. The need to simultaneously train numerous decision trees requires significant computational power. Additionally, while the model's predictions are often highly reliable, the aggregation of multiple decision trees makes it harder to interpret compared to simpler models. Despite these challenges, RF remains an outstanding choice in ML for its versatility, predictive accuracy, and effectiveness in sorting extensive data types (Lei et al., 2020), particularly in concrete compressive strength prediction.

2.4 Artificial neural network

Artificial Neural Networks (ANNs) represent computational frameworks inspired by the intricate biological structure of the human brain (McCulloch & Pitts, 1943). Their origins can be traced back to the 1940s when McCulloch and Pitts developed a network by combining several processing units (Walczak & Cerpa, 2003). Conceived as layers of interconnected nodes, or neurons, ANNs are specifically crafted to decipher intricate data relationships through a hierarchical abstraction process. Typically comprised of an input layer, one or more hidden layers, and an output layer, these neural networks are characterized by neurons interconnected across layers through weighted connections that undergo iterative refinement during the training phase to minimize prediction discrepancies. ANNs have demonstrated significant efficacy in delineating nonlinear patterns, thus proving invaluable in various domains such as predictive modeling of concrete compressive strength, where multifaceted and nonlinear correlations emerge among material attributes, curing conditions, and temporal aspects. Mathematically, an ANN is represented as a nonlinear regression model denoted by $f(x) = \varphi(w, x)$, where ϕ denotes a nonlinear model function, while perceptrons, the building blocks of ANNs, calculate the model function $f(x : w) = w^T x$ (Comito & Pizzuti, 2020). The algorithm achieves this by identifying a hyperplane that provides the best fit for the dataset while maximizing the separation margin for the support vectors.

ANNs are particularly suited for problems where the data has high dimensionality and non-linear dependencies. By analyzing interactions between the input features, an ANN learns patterns through repeated exposure to data. Training involves optimization methods such as backpropagation, which adjusts weights to reduce prediction errors. Enhancement methods such as gradient descent help refine the network's performance by minimizing a selected loss function. This iterative process enables the model to generalize effectively to new data, making it a versatile tool for modeling complex datasets. Through fine-tuning parameters like the number of hidden layers, neurons, activation functions, and learning rates, ANNs can achieve precise predictions, making them highly suitable for estimating concrete strength under diverse conditions (Walczak & Cerpa, 2003).

The flexibility of ANN architectures, such as deep learning models with multiple hidden layers, allows for the effective modelling of hierarchical data patterns. These networks can distill abstract representations from input datasets, thereby heightening their capability to discern nuanced interactions. Regularization techniques like dropout and early stopping effectively mitigate overfitting issues, ensuring robust model performance across varied datasets. Outcomes derived from ANN frameworks often exhibit a remarkable degree of accuracy, furnishing indispensable insights into the evolution of concrete strength, thereby consolidating their pivotal role in predictive modeling for concrete and other civil engineering applications.

2.5 Multilayer perceptron

Multilayer Perceptrons (MLPs), a subset of ANNs, was first utilized by Rumelhart & McClelland (1986) and is highly effective at uncovering intricate, arbitrary relationships within datasets. MLPs are composed of multiple layers are structured as a series of interconnected layers—input, hidden, and output—each comprising neurons that process information relayed from the preceding layer. This layered structure enables MLPs to capture and model complex patterns within data (Chan et al., 2023). In the context of predicting compressive strength, the MLP was chosen for its adeptness at capturing nuanced interactions among input variables, such as material composition and curing duration, which are inherently challenging to model using conventional linear methodologies. The potency of MLPs is particularly pronounced when dealing with data that exhibits non-linear characteristics, a common scenario in concrete-related predictions given the multitude of factors that can influence the material's strength.

The performance of the MLP model was enhanced through meticulous configuration of its architectural parameters, including the dimensions of the hidden layers and the number of neurons within each layer. The optimal design was ascertained through iterative experimentation, fine-tuning the depth (size of layers) and width (number of neurons per layer) to strike a balance between the model's capacity to learn and its ability to generalize. To mitigate overfitting and ensure robust performance with novel data, regularization techniques and cross-validation strategies, such as dropout and weight decay, were incorporated. By adopting this methodological approach, the MLP model was able to effectively encapsulate the intricate relationships within the dataset, yielding accurate predictions and providing valuable insights into the performance of concrete under varying conditions. The algorithm achieves this by identifying a hyperplane that provides the best fit for the dataset while maximizing the separation margin for the support vectors.

2.6 Support vector regressor

The Support Vector Regressor (SVR) is a sophisticated machine learning technique particularly effective for high-dimensional regression analysis. Derived from Cortes and Vapnik's (1995) foundational Support Vector Machine (SVM) architecture, SVR utilizes an epsilon-insensitive tube to handle continuous value prediction. This methodology constructs an optimal hyperplane that maximizes the margin around the target function while minimizing prediction errors—defined as deviations between experimental measurements and model outputs (Comito & Pizzuti, 2020).

Critical hyperparameters (C, Ɛ, and γ,) were calibrated via cross-validation to balance model complexity and generalization capacity. The regularization parameter C governs the trade-off between training error minimization and decision boundary smoothness. Epsilon (ε) establishes an error tolerance zone where minor deviations incur no penalty, while γ modulates the radius of influence for individual data points, with elevated values increasing model sensitivity to local variations. Through systematic parameter optimization, the SVR framework demonstrated robust predictive performance in compressive strength estimation, effectively mitigating overfitting risks while maintaining computational efficiency.

2.7 K-nearest neighbors

The K-Nearest Neighbors (KNN) algorithm, initially developed by Fix and Hodges (1989), represents a non-parametric machine learning method applicable to both classification and regression tasks. This instance-based learning approach determines output values for new observations through proximity analysis of K training samples, employing either majority voting (classification) or mean value calculation (regression). The algorithm's efficacy depends critically on two parameters: the neighborhood size (K) and the selected distance metric (typically Euclidean or Manhattan), which collectively govern prediction accuracy (Posma, 2019). As a lazy learning approach, KNN does not construct a model during the training phase but instead makes predictions by referencing stored data points during testing. Its simplicity, flexibility, and minimal assumptions about the data make it attractive for many applications, especially where the connections between input variables are intricate and non-linear (Kazemi et al., 2025). However, its performance is greatly influenced by the selection of K and the distance measure, necessitating careful optimization through methods like cross-validation. Additionally, the algorithm encounters computational difficulties, particularly when dealing with large datasets, as it requires calculating the distances from the test point to every training instance during the prediction (Islam et al., 2024b).

Despite its computational complexity, KNN remains effective for problems like concrete compressive strength prediction, where input variables exhibit intricate, local variations. KNN can adapt to these variations, providing accurate predictions even in noisy environments. In addressing computational challenges, advanced techniques such as KD-trees and Ball Trees are employed to accelerate nearest neighbor searches. Recent studies have proposed enhancements to KNN, including ensemble methods and attribute bagging, which reduce time complexity and improve prediction accuracy by narrowing the search space for nearest neighbors (Islam et al., 2024b). These extensions make KNN more efficient and scalable, particularly for big data applications. Moreover, KNN's ability to work with both categorical and continuous data, alongside its flexibility, continues to make it a valuable tool in various ML tasks, despite its need for optimization when dealing with large datasets.

2.8 Lasso regression

Lasso Regression, short for Least Absolute Shrinkage and Selection Operator, is a resilient linear regression method that serves dual purposes: it forecasts outcomes and facilitates feature selection and regularization, which are pivotal for enhancing model precision and comprehensibility. Initially presented by Tibshirani in 1996, LASSO employs L1 regularization, integrating a penalty term proportional to the absolute sum of the regression coefficients into the model's loss function. This penalty encourages certain coefficients to asymptotically approach zero, effectively eliminating irrelevant or less significant predictors from the model, thereby enhancing its predictive accuracy and interpretability (Tibshirani, 1996). This characteristic makes Lasso particularly advantageous when working with high-dimensional datasets containing numerous predictors, many of which may not substantially contribute to the predictive task.

The ability of Lasso to reduce the complexity of the model by zeroing out non-contributing features enhances its generalizability, mitigating overfitting while retaining significant predictive power. This ability to simplify the model while maintaining efficiency is especially beneficial in fields such as concrete compressive strength prediction, where numerous variables (e.g., cement, water-cement ratio, aggregate composition) influence the outcome. LASSO aids in the identification and retention of the most influential factors, such as cement content and water-cement ratio, while discarding less pertinent variables, resulting in more efficient and interpretable models (Satterfield et al., 2019).

In practice, Lasso's regularization parameter, often denoted as lambda (λ), controls the extent of penalization. Through techniques like cross-validation, this parameter is tuned to achieve a suitable equilibrium between bias and variance, guaranteeing the model avoids both overfitting and underfitting the data (Bühlmann & van de Geer, 2011). By incorporating this penalty, LASSO regression also functions as a mechanism for addressing multicollinearity, where predictors exhibit high correlation, thus enhancing the stability and robustness of the model. Consequently, LASSO is an invaluable technique in predictive modeling, especially when managing complex datasets where feature selection is crucial for extracting meaningful insights and augmenting model performance.

3 Literature review and problem statement

Accurately predicting the compressive strength of HPC is essential for optimizing mix designs and ensuring structural reliability. Over recent years, ML algorithms, particularly ensemble boosting and regression methods, have shown considerable promise in enhancing prediction accuracy and even most recently SHAP has been introduced to concrete technology and civil engineering. This section provides an overview of key contributions in the field and identifies areas that the current study aims to address.

Chou and Pham (2013) were among the frontrunners in applying ensemble learning techniques to predict the strength of HPC. Their pioneering research underscored the capacity of ensemble methods to effectively capture complex relationships inherent in concrete datasets, thus laying a strong foundation for subsequent scholarly inquiries. Subsequently, Erdal (2013) conducted a meticulous comparative analysis of various ensemble approaches, encompassing single, two-level, and hybrid models. The study concluded that decision tree-based ensembles, notably Random Forest Regression (RFR), showcased superior predictive performance compared to traditional decision tree models. This comparative analysis accentuated the potential of decision tree-based models in enhancing predictive accuracy, a finding that has since become a cornerstone for numerous studies in this domain.

Han et al., (2019) introduced a two-step ML framework incorporating RFR and variable importance measures. This innovative approach not only enhanced prediction accuracy but also improved model interpretability, providing deeper insights into the factors influencing compressive strength. Building on this, Kaloop et al. (2020) leveraged Gradient Boosting Machine (GBM) and the Adaptive Boosting (AdaBoost) algorithms to improve prediction accuracy, particularly in delineating the non-linear relationships prevalent in concrete data that often pose challenges to conventional models. Their research expanded the comprehension of how boosting techniques could be effectively employed to amplify prediction performance for concrete datasets, a notion that has since garnered considerable attention.

Nguyen et al. (2021a, 2021b) conducted a juxtaposition of different ML models and found that GBM and XGBoost provided the highest accuracy in forecasting both compressive and tensile strengths of HPC. Their study underscored the critical role of algorithm selection in ensuring dependable predictions for intricate concrete mixtures. Similarly, Lee et al. (2021) deployed the Categorical Boosting (CatBoost) algorithm to predict the strength of concrete-filled steel tubular structures, exemplifying the adaptability of ensemble methods in handling complex structural data. Collectively, these studies underscore the increasing interest in evaluating and optimizing the efficacy of boosting algorithms for concrete strength prediction.

Ahmad et al. (2022) made significant contributions by investigating the application of Decision Trees (DT) and AdaBoost in forecasting the compressive strength of fly ash-based geopolymer concrete. Their research underscored the efficacy of ensemble methods in handling unconventional concrete types, highlighting their potential to enhance strength prediction across a diverse range of concrete mixtures. Ekanayake et al. (2022) employed a combination of models, including XGBoost, Light Gradient Boosting Machine (LightGBM), and SHAP, to predict compressive strength, capturing a broader spectrum of concrete behavior. Their work demonstrated the flexibility of ensemble methods in addressing the multifaceted nature of concrete strength prediction. Additionally, Barkhordari et al. (2022) introduced the Super Learner (SL) algorithm, which merges multiple ensemble models to improve predictive accuracy. This innovative approach further validated the advantages of ensemble techniques in enhancing both model performance and reliability.

SHAP, a method introduced by Lundberg and Lee (2017), is extensively employed to enhance the interpretability of machine learning (ML) models by quantifying the contribution of each feature to the predictions. This method has been notably applied in civil engineering, particularly for forecasting the compressive strength of HPC. For instance, Islam et al. (2024a) utilized SHAP in conjunction with algorithms like Gradient Boosting, Random Forest, and CatBoost, in identifying key factors such as age, cement, and superplasticizer as primary contributors to HPC strength predictions. Similarly, Abdollahi et al. (2024) applied SHAP to road embankment safety factor predictions, highlighting the importance of parameters such as height, California bearing ratio, and cohesion. These studies underscore the utility of SHAP in providing interpretable insights into complex model predictions, thereby reinforcing its value in practical engineering applications where understanding the influence of each parameter is essential for optimizing mix designs and ensuring structural integrity.

While these studies collectively underscore the efficacy of ensemble and boosting techniques, a common limitation across the literature for predicting HPC compressive strength is the need for further optimization, particularly through hyperparameter tuning, to enhance model performance. Additionally, many studies have predominantly focused on traditional concrete mixtures, leaving ample room for further exploration of newer types of concrete, such as geopolymer or fly ash-based concretes, using ML techniques. The limited scope of datasets in some studies also presents an opportunity to explore a more diverse range of concrete mixtures, incorporating supplementary materials that contribute to strength enhancement, while leveraging SHAP to assess the individual significance and contributions of input variables—an innovative approach that provides a clearer understanding of how these parameters influence prediction outcomes.

The present study builds upon these advancements by exploring additional regression and boosting algorithms for predicting HPC compressive strength. A comparative evaluation and analysis will be conducted to assess the performance of these algorithms, with a specific focus on hyperparameter optimization. The objective is to identify the most effective models, ensuring their practical application in concrete mixture design and optimization. Furthermore, SHAP analysis will be utilized to assess the significance of each input feature, providing insights into how variables such as age, cement content, supplements, aggregates, and water-cement ratio influence compressive strength predictions. By addressing these gaps, this research aims to advance the ongoing development of precise and reliable ML models for predicting concrete strength, thereby offering valuable insights and support to both researchers and engineers in the civil engineering field.

4 Methodology

4.1 Data Source

The data used in this study was sourced from the UC Irvine ML Repository, specifically the 'Concrete Compressive Strength' dataset compiled by Professor I-Cheng Yeh of Chung-Hua University. The dataset consists of 1030 instances with nine attributes, including eight input variables and one output variable. This multivariate regression dataset aims to forecast concrete compressive strength based on its material composition and curing age.

4.2 ML model development

The selected ML models for predicting compressive strength include GBR, XGBoost, RF, SVR, ANN, MLP, Lasso Regression, and KNN. To ensure compatibility across these diverse models, a rigorous data preprocessing protocol was implemented, and hyperparameter optimization was conducted to maximize predictive accuracy. Cross-validation techniques were utilized to bolster the robustness of the models and mitigate the risk of overfitting. Each model was meticulously fine-tuned to capitalize on its inherent strengths. Also, SHAP analysis was implemented to examine the significance of the parameters on the output. The step-by-step methodology adopted for model development is as follows:

4.2.1 Data preprocessing

Data Cleaning The dataset was reviewed for type inconsistencies, anomalies, and missing values, though none were detected in this case.
Feature Scaling Using *StandardScale*, input features were standardized to achieve a mean of zero and a variance of one, ensuring uniformity across variables. Feature scaling is crucial for algorithms sensitive to magnitude, like SVR and neural networks (Kamolov, 2024).

4.2.2 Correlation analysis

To assess the linear and non-linear relationships between input features and to detect potential multicollinearity, a correlation analysis was conducted. A correlation heatmap was employed to visually represent these relationships. Pearson and Kendall's tau correlation coefficients were utilized to quantify both linear and non-linear associations among the variables.

4.3 Pearson correlation coefficient formula

Pearson's correlation coefficient (r) is obtained using:

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{\left[n\Sigma x^2 - (\Sigma x)^2\right]\left[n\Sigma y^2 - \left(\Sigma y\right)^2\right]}}$$

Here, n represents the total observations, while x and y represent the variables being compared. The sum of the products of the paired observations is expressed as $\sum xy$, and $\sum x$ and $\sum y$ refer to the sums of the individual observations. Additionally, $\sum x^2$ and $\sum y^2$ represent the sums of the squares of the individual observations respectively.

4.4 Kendall's Tau correlation coefficient formula

Kendall's tau (τ) measures the strength and direction of the non-linear association between the variables. It is computed using the following formula:

$$\tau = \frac{C - D}{\sqrt{(C + D + T_x)(C + D + T_y)}}$$

where C represents the count of concordant pairs, which are pairs where the rank order of both variables is consistent, and D denotes the count of discordant pairs, where the rank order of the variables differs. Additionally, T_x and T_y refer to the number of tied ranks within each of the variables (x and y), respectively.

4.4.1 Splitting the dataset

The dataset was partitioned into predictors ($\boldsymbol{X}$) and the target variable ($\boldsymbol{y}$). A train-to-test ratio of 90:10 was used, with a fixed *random_state* of 0 to ensure reproducibility. The train subset was used for model training, while the test subset was reserved for assessing the model's performance.

4.4.2 Model training

Each selected ML algorithm was trained using python (v3.13.0), pip (v23.3.1) and relevant libraries such as scikit-learn (v1.5.2), NumPy (v2.1.3), SciPy (v1.14.1), pandas (v2.2.3), matplotlib (v3.9.3), and seaborn (v0.13.2). The training regimen encompassed the fitting of these algorithms to the designated training dataset, subsequent to which a meticulous hyperparameter optimization process was initiated to augment the models' predictive efficacy.

4.4.3 Hyperparameter tuning and cross validation

Extends the existing body of knowledge by employing a comprehensive framework for hyperparameter tuning and cross-validation across a diverse set of eight ML algorithms, thereby ensuring the robustness of the models' performance. The hyperparameter optimization was executed via the GS method, which systematically traversed a broad spectrum of hyperparameter values to ascertain the most optimal configurations. To bolster the reliability of the models, a k-fold cross-validation strategy was implemented. This involved partitioning the training dataset into multiple distinct subsets, or folds, which served to counteract the potential for overfitting and to affirm the models' generalizability, thereby ensuring consistent performance across various data partitions (Rathakrishnan et al., 2022).

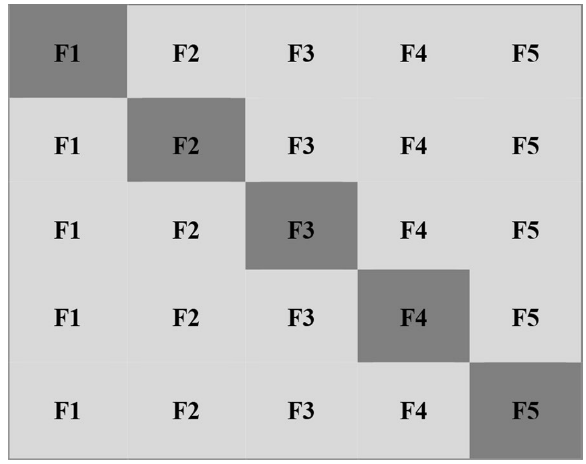

Fig. 1 Schematic of the cross-validation method

Figure 1 illustrates the k-fold cross-validation approach adopted in this study to enhance the reliability and robustness of the ML models. The dataset was partitioned into five different folds (n_splits = 5), with each segment taking turns as the validation set while the others were utilized for training. This iterative process was applied to all algorithms, enabling a comprehensive evaluation of their performance. By averaging results across all folds, the approach inhibited overfitting and delivered a more precise assessment of accuracy and generalizability. Consequently, this approach ensured a consistent performance regardless of varying data distributions.

The selection of hyperparameter ranges delineated in Table 1 was predicated on a synthesis of extant scholarly research and preliminary empirical trials, with the dual objectives of optimizing computational expedience and enhancing the predictive efficacy of the models. Nguyen et al., (2021a, 2021b) provide robust benchmarks for selecting hyperparameter ranges for algorithms such as XGBoost, GBR, and SVR. The ranges for n_estimators, learning_rate, and max_depth for XGBoost and GBR closely align with those used by Nguyen et al., where optimal results were achieved with values like n_estimators between 100 and 300, learning_rate between 0.01 and 0.2, and max_depth up to 7. Similarly, for SVR, the ranges for C, epsilon, and kernel reflect settings that have

Table 1 Hyperparameter Values

Model	Parameter Grid [Value]
XGBoost	'n_estimator': [100, 150, 200] 'learning_rate' [0.01, 0.1, 0.2] 'max_depth': [3, 5, 7] 'subsample': [0.8, 1.0]
GBR	'n_estimator': [100, 150, 200] 'learning_rate' [0.01, 0.1, 0.2] 'max_depth': [3, 5, 7] 'subsample': [0.8, 1.0] 'min_sample_split': [2, 5] 'min_sample_leaf': [1, 3]
RF	'n_estimator': [100, 200, 300] 'learning_rate' [0.01, 0.1, 0.2] 'max_depth': [None, 10, 20, 30] 'min_sample_split': [2, 5, 10] 'min_sample_leaf': [1, 2, 4] 'max_features': ['sqrt', 'log2']
SVR	'C': [0.1, 1, 10] 'epsilon': [0.01, 0.1, 0.2] 'kernel': ['linear', 'rbf']
MLP	'hidden_layer_sizes': [(100,), (150,)] 'activation': ['relu','tanh'] 'solver': ['adam'] 'learning_rate': ['constant', 'adaptive'] 'alpha': [0.0001, 0.001, 0.01]
ANN	'hidden_layer_sizes': [(50,) (100,), (100), (150)] 'activation': ['relu', 'tanh'] 'learning_rate_init': [0.001, 0.01, 0.1] 'alpha': [0.0001, 0.001, 0.01]
Lasso	'alpha': np.logspace(-3, 2, 20) 'max_iter': [10000]
KNN	'n_neigbours': range (1, 21) 'weights': ['uniform', 'distance'] 'p': [1, 2]

been shown to improve prediction accuracy for concrete compressive strength in prior research.

For neural network-based models like MLP and ANN, the hyperparameter ranges, including hidden_layer_sizes, activation, and alpha, were informed by guidelines from Silva et al. (2023) and Terlumun et al. (2024). These studies emphasize the importance of balancing underfitting and overfitting through controlled variations in the number of neurons and regularization parameters (alpha). For instance, lower alpha values were scrutinized to ascertain adequate generalizability, while hidden_layer_sizes were calibrated in accordance with prevalent configurations for datasets of modest to intermediate scale.

In addition, hyperparameter values for algorithms like RF and KNN were informed by standard practices in ML, as documented in foundational studies. For RF, parameters like max_depth, min_samples_split, and max_features were varied to capture optimal tree structures, while for KNN, n_neighbors and weights were tuned to achieve balanced performance across different distance metrics.

The parameter grid was designed to ensure coverage of a broad range of values while incorporating insights from literature. Cross-validation was employed to identify the best-performing hyperparameter combinations, reducing overfitting and ensuring reproducibility. The ranges selected ultimately allowed for a systematic exploration of the parameter space, yielding optimal configurations for predicting HPC compressive strength.

The parameter grids tested are detailed in Table 1, showcasing key parameters such as *n_estimators*, *learning_rate*, and *max_depth* for models like XGBoost, GBR, and RF. Additionally, specific hyperparameters were explored for other models: *C*, ε, and γ for SVR; *hidden_layer_sizes*, *activation*, *solver*, *learning_rate*, and *alpha* for MLP; *hidden_layer_sizes*, *activation*, *learning_rate_init*, and *alpha* for ANN; *alpha* and *max_iter* for Lasso; and *n_neighbors*, *weights*, and *p* for KNN.

4.4.4 Evaluating model performance (Accuracy & Efficiency Metrics)

Model evaluation was conducted utilizing the test dataset to assess the accuracy and error levels of various models. The evaluation entailed the use of several metrics, including R^2, MSE, RMSE, MAE, and MAPE. Following this assessment, a comparative analysis was executed to identify the models exhibiting superior performance based on these evaluation criteria. Furthermore, computational cost was factored in through the consideration of training time to provide a comprehensive evaluation of the models' efficiency. This approach ensured that the selection of the most appropriate model took into account both accuracy and efficiency. Besides prediction accuracy, the efficiency of each algorithm was appraised, with training time serving as a critical measure to gauge the practical feasibility of implementing these models in real-world scenarios. Training time was measured by recording the duration required to train the model, calculated as $T = t_e\text{-}t_s$, where t_e and t_s denote the timestamps marking the end and start of the training process, respectively.

4.4.5 SHAP analysis (Feature Importance)

Integrating SHAP with ML models provides a powerful technique for understanding how different input features influence predictions. This method enables a comprehensive evaluation of the contribution of each variable to the model's output. To investigate feature impact, SHAP was employed, offering deeper insights into how each parameter affects HPC strength predictions. The TreeExplainer function in SHAP was specifically used to estimate these contributions by computing SHAP values, which provide an approximation of each feature's effect on model predictions. These values, formulated through an additive attribution framework, are mathematically represented by the equation (Huang et al., 2020):

$$SHAP\_value\_of\_feature = g(z') = \emptyset_0 + \sum_{j=1}^{M} z'_j \emptyset_j$$

where z' represents the "Coalition vector" and g is the explanation model. This approach allows for a clear understanding of how each feature influences the model's prediction for a specific instance (Keleko et al., 2022).

To ensure that the SHAP analysis reflects the model's performance across different data splits, the feature importance assessment was conducted using the k-fold cross-validation approach (with n_splits = 5). Specifically, for each of the fold, the model was trained on the training dataset. The SHAP values were then computed for each fold, and the average SHAP values across all folds were used to determine the overall impact of each feature. By employing k-fold cross-validation, the analysis mitigates potential biases that could arise from relying on a single train-test split, providing a comprehensive view of how each feature influences the model's predictions across different subsets of the data. This method ensures that the feature importance evaluation is robust, offering insights into the stability and consistency of feature effects on HPC strength predictions.

4.4.6 Accepting user input

The final step in the predictive modeling process involves the facilitation of user input for predictive purposes. Users are empowered to input values for each pertinent feature, which are subsequently standardized and processed in accordance with the operational requirements of the trained model. This functionality empowers the model to deliver real-time predictions predicated on the user-input data, thereby ensuring the provision of precise and immediate results.

Various modelling techniques can aid in comprehending intricate data interactions and in determining the most suitable model for forecasting strength in HPC. This method guarantees a thorough analysis of the issue at hand and utilizes the advantages of different ML methods. The code repository built for this study can be found at SamML.

5 Results and discussions

5.1 ML model performance comparison

The results section offers an in-depth assessment of the ML models applied in predicting the compressive strength of HPC. A thorough analysis is presented, showcasing how each model performed in terms of both accuracy and error. For clarity, a juxtaposition of the models' outcomes was conducted, allowing for a direct assessment of their relative performance. In Table 2, the values of the performance metrics for each algorithm have been

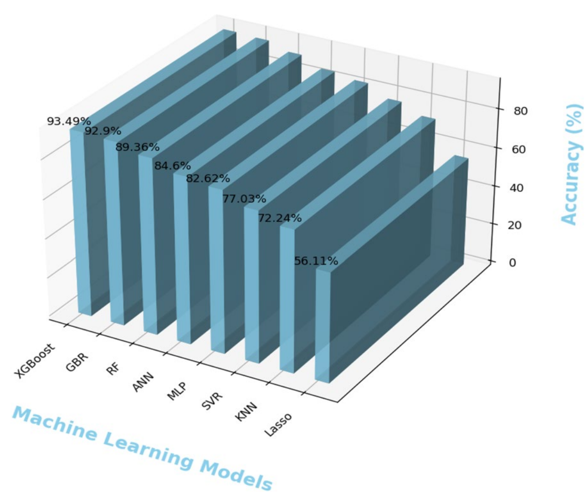

Fig. 2 ML Accuracy Chart

Table 2 Accuracy Metrics and Error Values for ML Models on test data

Algorithms					Evaluation Indicators	
	Type	Data Volume	R^2	MSE	MAE	RMSE
XGBoost	Ensemble	1030	0.9349	12.59	2.498	3.55
GBR	Ensemble	1030	0.9209	15.30	2.663	3.91
RF	Ensemble	1030	0.8936	27.59	3.798	5.25
ANN	Individual	1030	0.8460	29.79	3.835	5.46
MLP	Individual	1030	0.8262	33.61	4.381	5.80
SVR	Individual	1030	0.7703	44.43	4.902	6.67
KNN	Individual	1030	0.7224	71.96	6.185	8.48
Lasso	Individual	1030	0.5611	113.77	8.637	10.67

meticulously provided, including various measures of predictive accuracy and the corresponding error values. These values, derived from the final model evaluation on the test set, represent the predictive performance on unseen data. This comprehensive overview not only illustrates the predictive capabilities of each model but also provides a better understanding of their strengths and weaknesses in modeling HPC.

As presented in Table 2 and Fig. 2, ensemble learning algorithms, particularly XGBoost, GBR, and RF, achieved superior predictive accuracy for concrete compressive strength estimation. XGBoost demonstrated optimal performance with an R^2 of 0.9349, capturing 93.49% of data variance, and exhibited the lowest error metrics (MSE = 12.59, MAE = 2.498, RMSE = 3.55). GBR closely followed with comparable accuracy (R^2 = 0.9209, MSE = 15.30, RMSE = 3.91), effectively modeling nonlinear material relationships. RF maintained strong predictive capability (R^2 = 0.8936, MSE = 27.59), outperforming all non-ensemble counterparts as illustrated in Fig. 2. These results confirm ensemble methods' effectiveness in balancing predictive precision and error minimization for complex regression tasks.

Neural network models such as ANN and MLP exhibited moderate performance, with ANN achieving R^2 = 0.8460 (MSE = 29.79, RMSE = 5.46) and MLP attaining R^2 = 0.8262 (MSE = 33.61, RMSE = 5.80). While demonstrating baseline competency, their elevated error margins suggest limitations in capturing intricate parameter interactions without extensive hyperparameter optimization, as evidenced by the comparative error distributions in Fig. 2. The relatively higher error values for ANN and MLP indicate that these models experienced slight difficulty with capturing intricate interactions within the dataset, necessitating additional enhancements in their architecture or feature engineering.

On the other hand, SVR, KNN, and Lasso displayed comparatively reduced efficacy. SVR yielded R^2 = 0.7703 (MSE = 44.43, RMSE = 6.67), followed by KNN (R^2 = 0.7224, MSE = 71.96, RMSE = 8.48) and Lasso (R^2 = 0.5611, MSE = 113.77, RMSE = 10.67).As visualized in Fig. 2, the disparities in accuracy metrics among these models underline the superior generalization capabilities of ensemble algorithms over individual approaches like SVR, KNN, and Lasso.

Overall, the analysis of Table 2 and Fig. 2 underscores the importance of selecting robust ML models for making HPC compressive strength predictions. The ensemble methods, particularly XGBoost and GBR, stand out for their high accuracy and low error rates, making them ideal for complex regression tasks. Neural networks like ANN and MLP, while moderately effective, require further tuning to improve their predictive performance. Individual models such as SVR, KNN, and Lasso lag in both accuracy and consistency, reinforcing the value of ensemble techniques for handling intricate datasets. This comprehensive evaluation highlights the distinct advantages of advanced algorithms in achieving reliable and precise predictions in civil engineering applications.

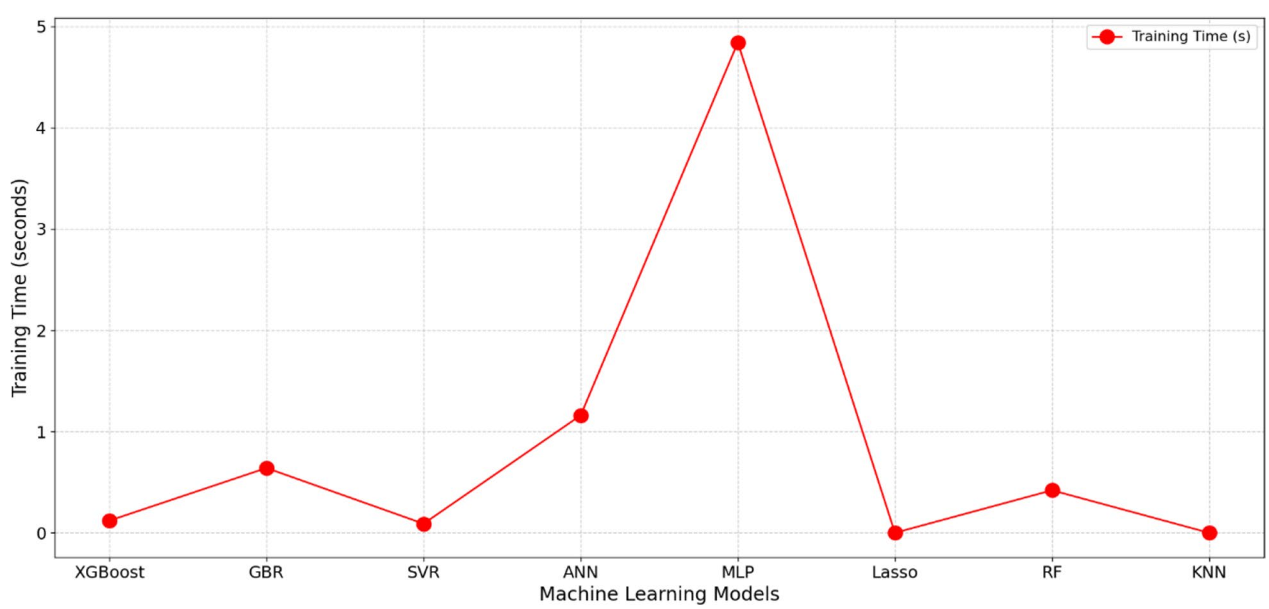

Fig. 3 Line chart representing the ML training time

5.2 Computational efficiency of ML models

Figure 3 presents the training time for the various ML models, showcasing a relative representation of their computational cost. The red dots denote the training time, with MLP taking the longest at 4.64 s, followed by ANN at 1.16 s. These values highlight the higher computational demands of models with complex architectures. GBR, with a training time of 0.84 s, also reflects a moderate computational cost, making it reasonable for scenarios where slightly higher training time is acceptable. In contrast, SVR and RF demonstrate faster training times of 0.09 and 0.42 s, respectively, while XGBoost stands out with an impressive 0.12 s, combining efficiency with robust predictive performance. Lasso and KNN, both showing 0.00 s, indicate negligible training times, as their methodologies do not involve iterative parameter adjustments during training.

The concept of training time serves as a vital parameter in evaluating computational cost, especially in scenarios involving extensive datasets or applications with time constraints. Models such as XGBoost exemplify a notable equilibrium between efficiency in training duration and precision in results, rendering them particularly advantageous in such contexts. The efficiency of Lasso model is inherently derived from its simplistic linear formulation, requiring minimal computational resources. Conversely, KNN's zero training time results from its instance-based approach, where the computational load is deferred to prediction rather than training. These characteristics make Lasso and KNN highly efficient for datasets where speed is prioritized over accuracy. Overall, XGBoost emerges as a standout performer due to its ability to maintain excellent accuracy alongside a remarkably low training time, making it ideal for scenarios demanding both precision and efficiency.

5.3 Correlation of variables

5.3.1 Using Pearson's correlation coefficient

In this section, we undertake a comprehensive correlation analysis of all parameters, encompassing both input variables pertaining to concrete composition and the output variable, namely concrete compressive strength. The correlation heatmap presented in Fig. 4 offers a visual representation of the relationships between input and

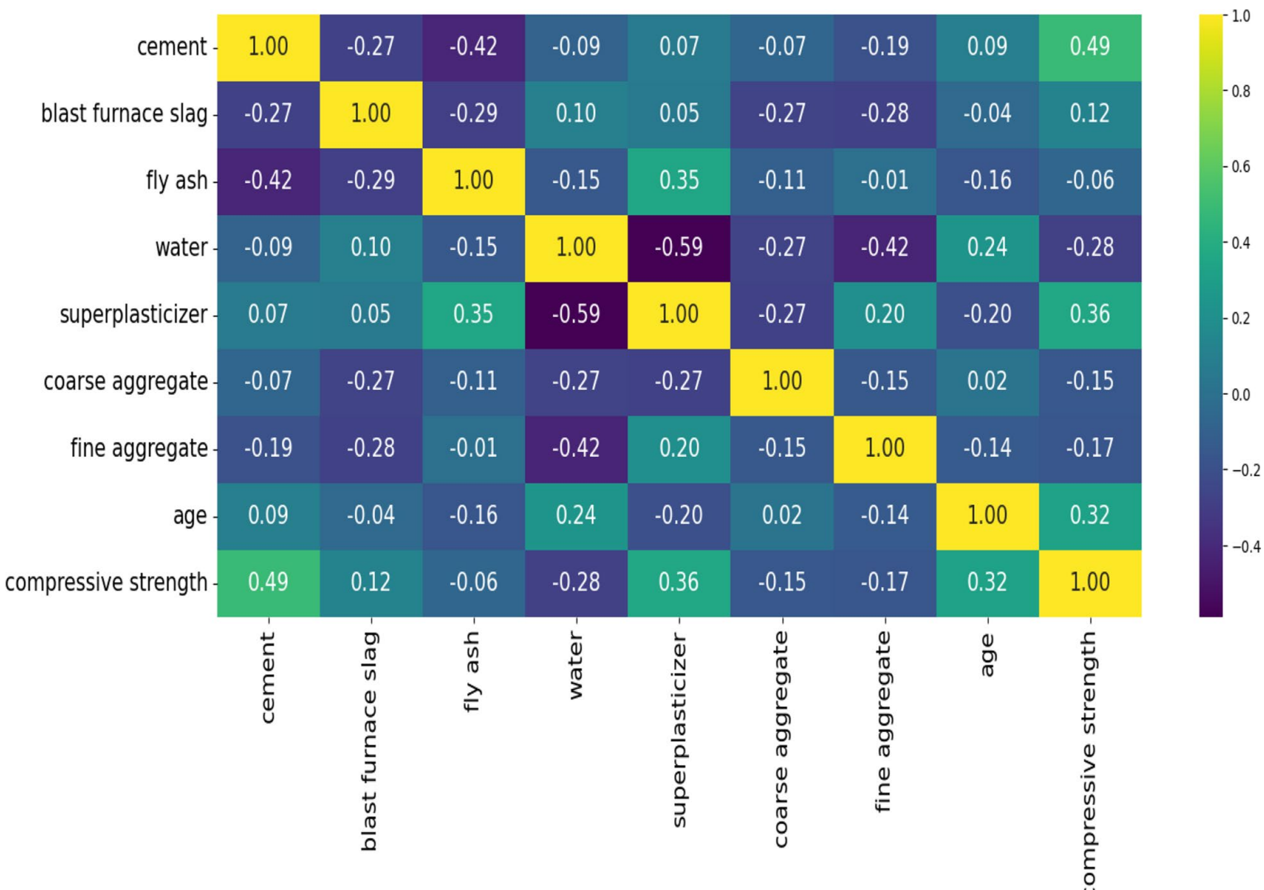

Fig. 4 Pearson's Correlation Heatmap

output variables. Following a thorough examination, it has been determined that specific variables demonstrate a more pronounced correlation with concrete strength than others.

For instance, an analysis of the data reveals a moderately positive correlation between cement content and the coefficient of determination, r2, which is 0.49. This finding indicates that variations in cement content have a substantial impact on concrete strength, with higher cement content typically resulting in increased strength. Conversely, the correlation between the content of blast furnace slag and the compressive strength of concrete is comparatively negligible, with a correlation coefficient of 0.12. This finding suggests that alterations in the composition of blast furnace slag have a marginal effect on the strength of concrete when compared to other factors considered.

Similarly, fly ash content shows a weak negative correlation (-0.06) with concrete compressive strength, suggesting that higher levels of fly ash may marginally decrease strength. Water content exhibits a moderate negative correlation (-0.28) with concrete strength, indicating that excessive water content can lead to reduced strength due to increased porosity and decreased hydration rates.

The presence of superplasticizer displays a moderate positive correlation (0.36) with concrete compressive strength, implying that its addition may enhance the material's strength properties. Coarse aggregate and fine aggregate demonstrate weak negative correlations (-0.15 and -0.17, respectively) with concrete strength, indicating minor impacts compared to other variables.

Furthermore, the age of concrete exhibits a moderate positive correlation (0.32) with compressive strength, highlighting the significance and importance of curing time in achieving optimal strength characteristics. Finally, the perfect correlation coefficient of 1.00 between concrete compressive strength and itself validates the analysis methodology and serves as a reference point for comparison.

5.3.2 Using Kendall's tau correlation matrix

The Kendall's Tau correlation matrix, as visualized in Fig. 5, provides insights into the nonlinear interactions among input variables and their relationship with compressive strength. The heatmap uses a color gradient to represent the strength and direction of correlations, with the darker blue shades indicating negative correlations and lighter shades signifying positive correlations. Annotated numerical values enhance clarity, allowing for an

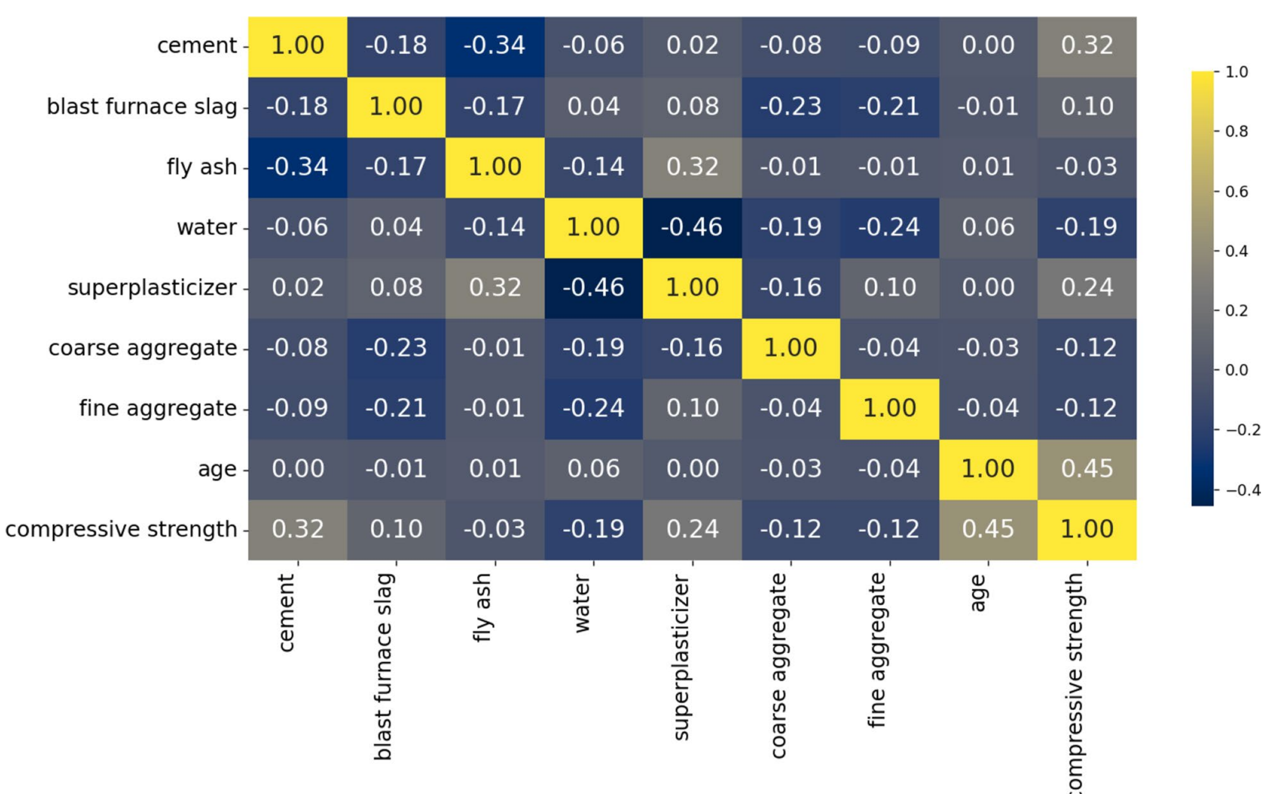

Fig. 5 Kendall's tau nonlinear interaction Correlation Heatmap

intuitive understanding of the relationships between variables. The diagonal elements with yellow shades, showing perfect correlations (1.00), serve as a reference, while off-diagonal elements reveal the strength and direction of pairwise interactions.

Table 3 Quantitative properties of variables

Variables	Mean	Range	SD	Type
cem (kg/m^3)	281.2	102.0—540.0	104.5	Independent
bfs (kg/m^3)	73.9	0.0—359.4	86.2	Independent
fa (kg/m^3)	54.2	0.0—200.1	64.0	Independent
Wtr (kg/m^3)	181.6	121.8—247.0	21.3	Independent
sp (kg/m^3)	6.2	0.0—32.2	6.0	Independent
cag (kg/m^3)	972.9	801.0—1145.0	77.7	Independent
fag (kg/m^3)	773.6	594.0—992.6	80.1	Independent
Age $(days)$	45.7	1—365	63.1	Independent
cs (MPa)	35.8	2.3—82.6	16.7	Dependent

Key relationships emerge from the matrix. Cement exhibits a moderately positive correlation with compressive strength (0.32), indicating its significant role in strength development. Age shows the highest positive correlation (0.45), emphasizing the critical influence of curing time on strength. Conversely, water has a negative correlation with compressive strength (-0.19), reflecting the detrimental impact of higher water content on concrete strength due to increased porosity. The superplasticizer shows a moderate positive correlation (0.24), highlighting its effectiveness in improving workability and reducing water content without compromising strength.

The interactions among the variables in concrete mixtures offer valuable insights into their collective impact on compressive strength. Fly ash shows a negative correlation with cement (– 0.34) and water (– 0.14) but a positive correlation with the superplasticizer (0.32), reflecting its role as a supplementary cementitious material. Coarse and fine aggregates exhibit weak negative correlations with compressive strength, both at

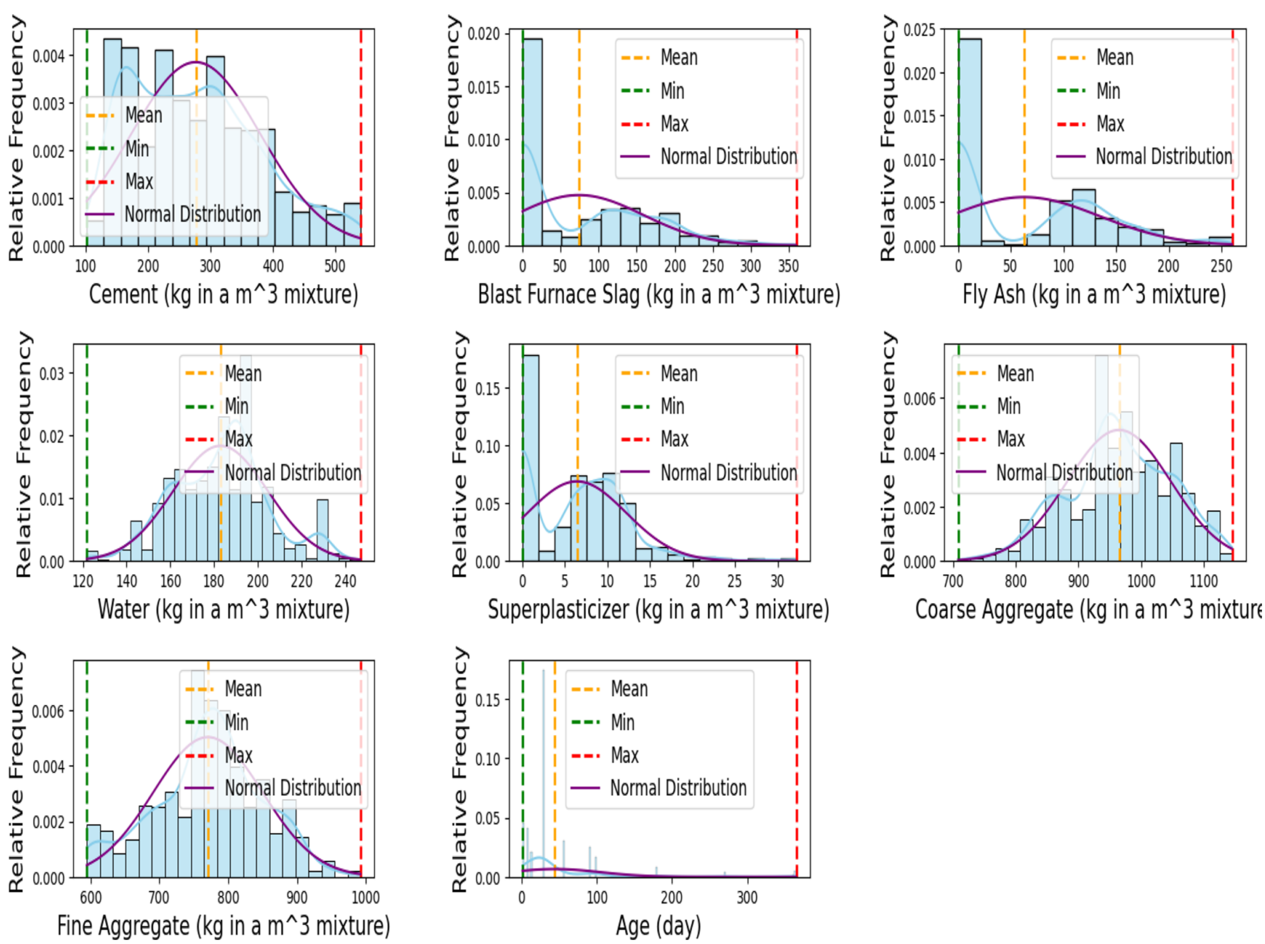

Fig. 6 Histogram showing Statistical analysis of Input parameters

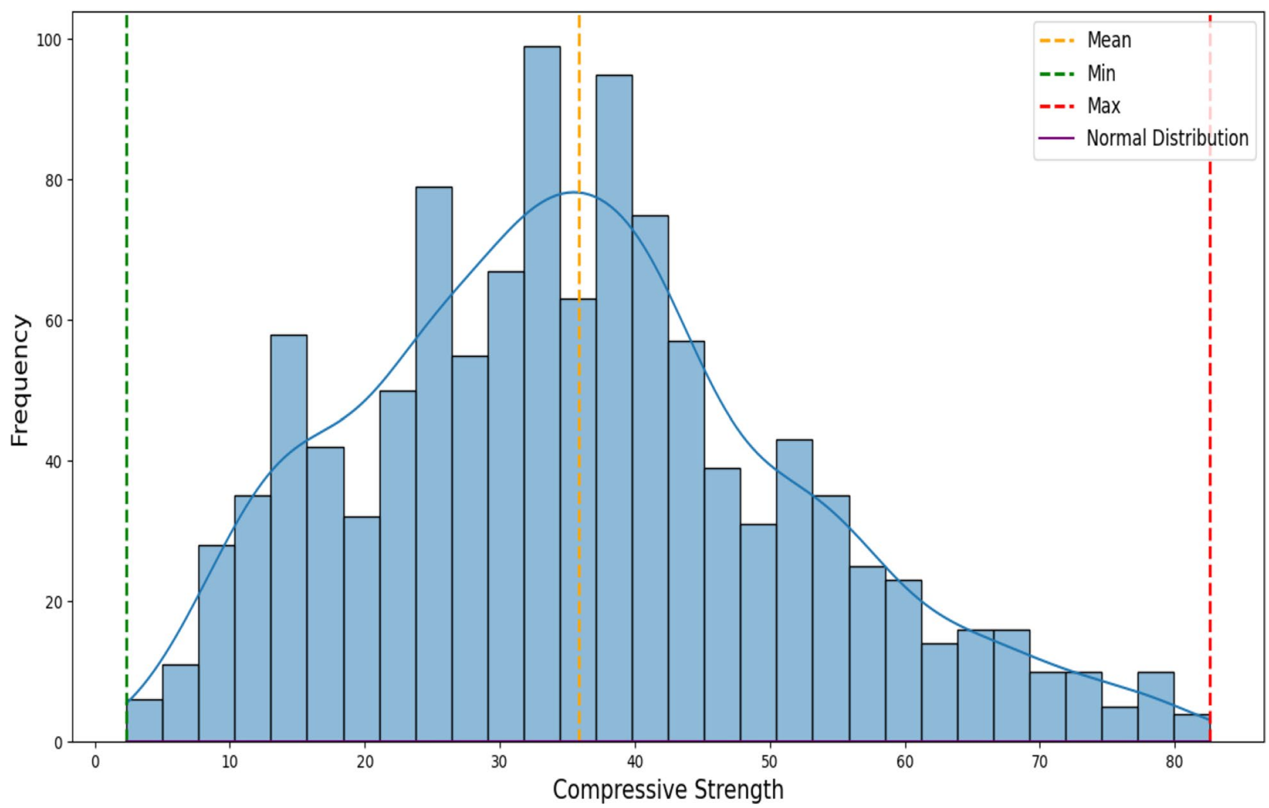

Fig. 7 Histogram illustrating the statistical analysis of output variables

– 0.12, indicating their relatively lesser direct impact. The heatmap further underscores the importance of cement, age, and superplasticizer in enhancing compressive strength while identifying trade-offs, such as the adverse effects of excessive water content.

5.4 Numerical characteristics of the parameters/variables

The numerical characteristics of the parameters, as summarized in Table 3 and illustrated in Figs. 6 and 7, provide a detailed understanding of the dataset's variability and distribution patterns. The relative frequency depicted in these figures represents how frequently each data point occurs, highlighting the reproducibility and consistency of the dataset. Cement content, a critical component in concrete mix design, ranges from 102 to 540 kg/m^3, with a mean of 276.50 kg/m^3 and a standard deviation of 103.47 kg/m^3. This wide range reflects the adaptability of cement quantities in achieving specific strength and durability requirements. Supplementary materials, such as blast furnace slag and fly ash, exhibit considerable variability, with slag ranging from 0 to 359.40 kg/m^3 (mean: 74.27 kg/m^3) and fly ash from 0 to 260.00 kg/m^3 (mean: 62.81 kg/m^3), demonstrating their selective use to optimize concrete performance. Water content, with a relatively narrow range of 121.75 to 247.00 kg/m^3 (mean: 182.98 kg/m^3), and superplasticizer, ranging from 0 to 32.20 kg/m^3 (mean: 6.42 kg/m^3), highlight the precision required in their proportions to ensure workability and strength.

Coarse aggregate content spans from 708 to 1145 kg/m^3 (mean: 964.83 kg/m^3), while fine aggregate ranges from 594 to 992.60 kg/m^3 (mean: 770.49 kg/m^3), reflecting diverse gradations that contribute to the structural integrity of the concrete. The age parameter, a critical factor in strength development, varies significantly from 1 to 365 days, with a median value of 28 days, emphasizing the dataset's coverage of both early and long-term curing periods. Compressive strength, the output variable, ranges from 2.49 to 14.13 MPa, with a mean of 6.76 MPa, capturing the diverse performance of concrete mixes under varying conditions.

Figure 6 presents histograms for each input variable, with blue curves superimposed to represent the fitted normal distribution. The blue curves underscore the central tendency and variability of the data, offering valuable insights into its distribution characteristics. The y-axis, representing relative frequency, normalizes the data to indicate the proportion of observations within each bin relative to the total dataset. This visualization helps assess the consistency of the data and how closely it follows a normal distribution. The input variables are well-distributed, suggesting that the dataset is effectively normalized and provides a balanced representation of the data.

Similarly, Fig. 7 depicts the histogram of compressive strength values, where the blue normal distribution curve highlights the concentration of values around the mean of 6.76 MPa. The curve provides a visual approximation of the dataset's distribution, helping to identify patterns such as skewness or deviations from normality. The relative frequency here serves the same purpose, normalizing the data to facilitate comparison and interpretation.

By integrating these visualizations with the summary statistics presented in Table 3, the analysis delivers a holistic understanding of the dataset's variability, central tendencies, and distribution patterns. The incorporation of normal distribution curves situates the data within a statistical framework, providing insights into the extent of variability and the congruence of parameters with anticipated distributions. These figures and their accompanying explanations address the reviewer's concerns by elucidating the significance of the blue curves and the concept of relative frequency within the histograms.

5.5 Relative importance of input parameters

Figure 8 presents a comparative analysis of the relationship between various independent (input) parameters and their influence on compressive strength, a pivotal factor in the refinement of concrete mixtures and the assurance of precise performance predictions. Cement is identified as the predominant factor affecting compressive strength, highlighting the criticality of considering

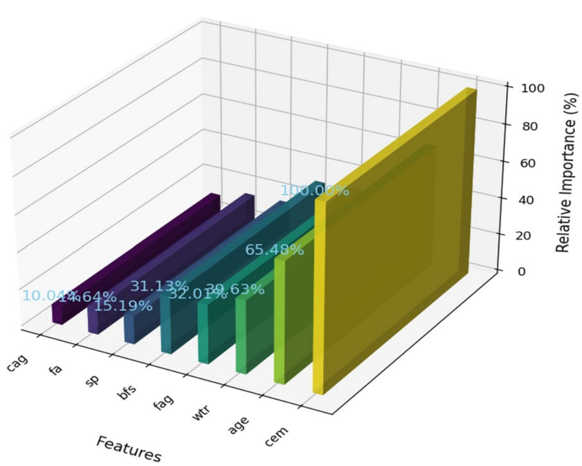

Fig. 8 Relative importance of input parameters on compressive strength

curing duration and hydration kinetics. Age is a close second, underscoring the imperative for meticulous control during the mixing process to optimize strength properties, in line with established engineering practices in concrete strength assessment.

Moreover, water content, superplasticizer dosage, and supplementary materials like fly ash and blast furnace slag also play significant roles in determining HPC's compressive strength. Their contributions are intricately linked to factors like workability and pore structure development, highlighting the intricate nature of concrete performance optimization. This underscores the importance of a comprehensive approach in mixture design, where each parameter's impact is carefully assessed and balanced to achieve desired performance outcomes. The observations depicted in Fig. 8 are corroborated by the findings from the correlation heatmaps in Figs. 4 and 5, which further substantiates the significance of cement and age as key determinants of compressive strength.

5.6 SHAP analysis: impact of input parameters on compressive strength

The SHAP analysis depicted in Figs. 9 and 10 provides an in-depth evaluation of the contribution of each input variable to the prediction of compressive strength, with the significance of each variable quantified by the mean SHAP values. Age emerges as the most influential factor, with a mean SHAP value of +8.18, underscoring its pivotal role in determining the strength of HPC. The positive SHAP value for Age suggests that the longer the concrete cures, the greater its compressive strength becomes, even surpassing the anticipated strength. This underscores the necessity of incorporating curing time into the mix design, as concrete's strength continues to accrue over time, with extended curing periods leading to enhanced compressive strength.

Cement is the second most influential factor, with a mean SHAP value of +6.92, reinforcing its crucial role in the hydration process and its direct bearing on the compressive strength of concrete. The presence of cement catalyzes the chemical reactions that lead to strength development, and its positive SHAP value indicates that an increase in cement content positively impacts the strength of the mixture. While cement is indispensable, the data indicates that Age exerts an even more profound influence on the ultimate compressive strength, especially as the concrete matures beyond its initial curing phase.

Water, with a mean value of +4.76, also plays a significant role in compressive strength, though its impact is more nuanced. Water is essential for hydration, but an optimal balance must be maintained. Excessive

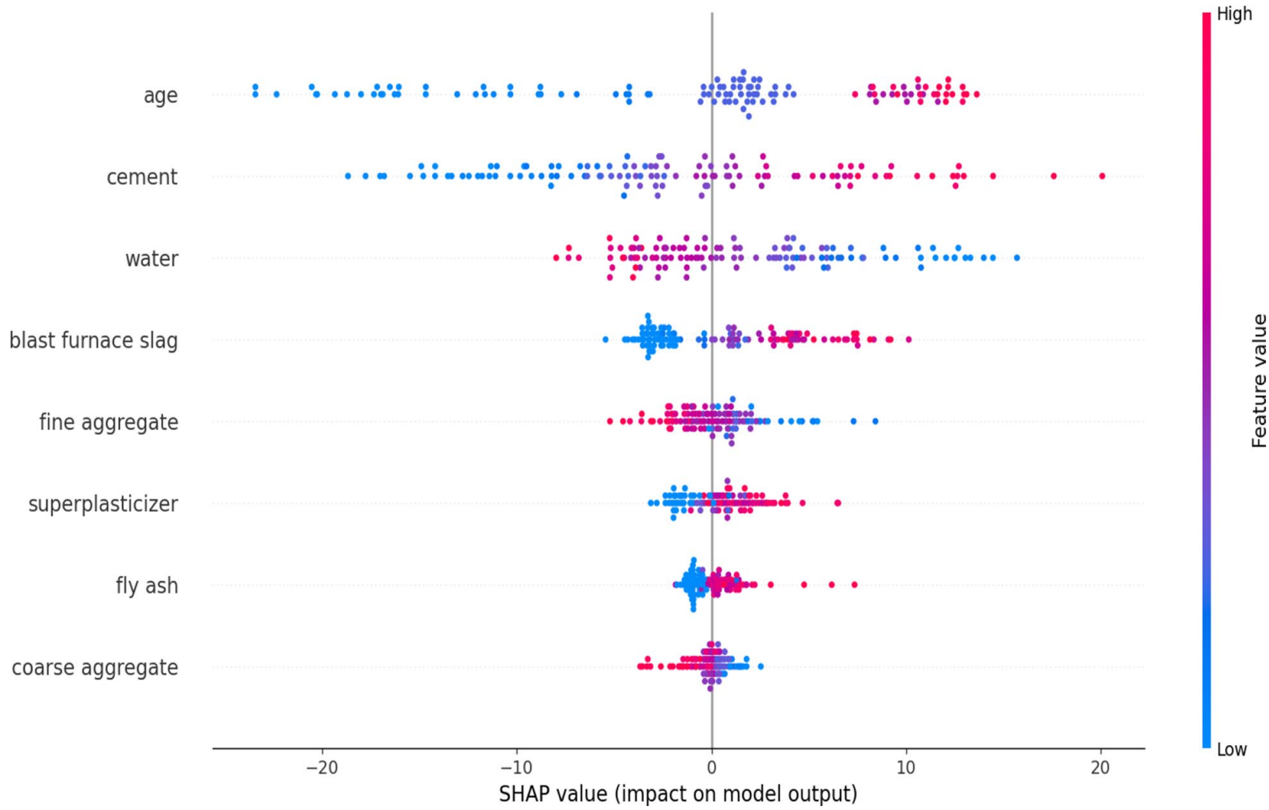

Fig. 9 SHAP analysis of feature Input

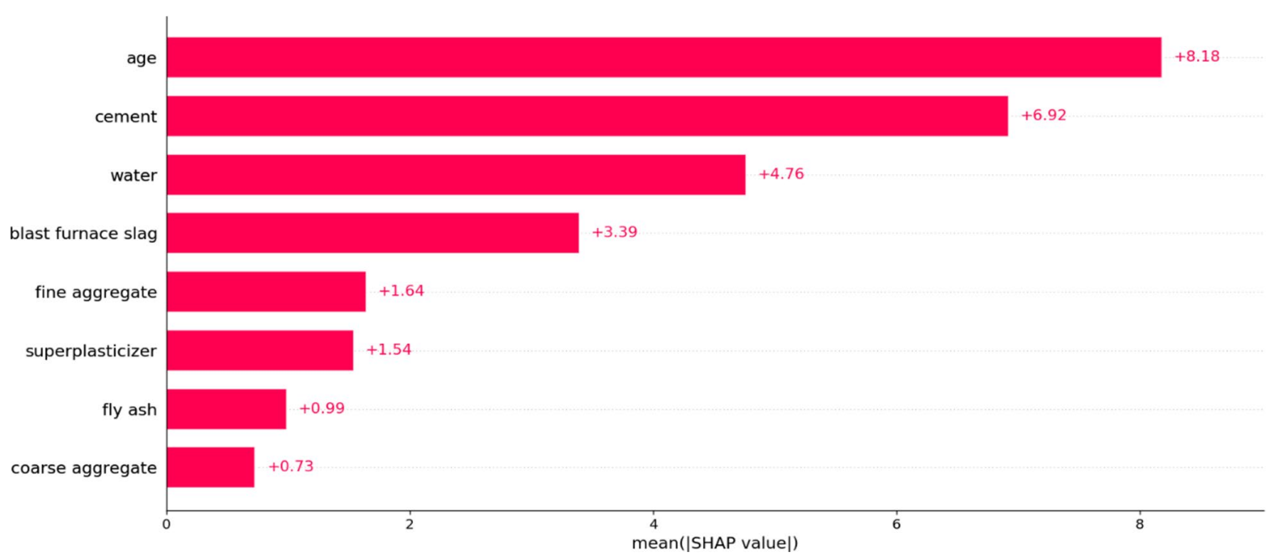

Fig. 10 Bar chart showing Mean SHAP Values

water can dilute the cement content, leading to a decrease in strength. However, within a certain range, Water contributes positively to the hydration process, enhancing strength. Fine aggregate, with a mean SHAP value of + 1.64, shows a moderate positive contribution to the overall strength. The amount of fine aggregate influences the concrete's workability and strength, but its effect is secondary to the Cement and Age.

Supplementary materials such as Blast Furnace Slag, Superplasticizer and Fly Ash also contribute to the strength prediction, but their impact is less pronounced. Blast Furnace Slag, with a mean SHAP value of + 3.39, has a moderate positive effect, indicating that its inclusion enhances the compressive strength to some degree. Superplasticizer, with a mean SHAP value of + 1.54, and Fly Ash, with + 0.99, exhibit more subtle influences, suggesting that while they improve the concrete's workability and durability, their impact on compressive strength is less significant compared to Cement or Age. Coarse Aggregate, with a mean SHAP value of + 0.73, has the least contribution, reflecting its role in providing bulk to the mixture, but with a lesser influence on HPC compressive strength compared to other factors.

The SHAP analysis highlights the intricate interplay among these variables, offering valuable insights for the optimization of concrete mixture designs. By employing k-fold cross-validation to derive the SHAP values, the analysis ensures a robust and generalized understanding of the features' impact, safeguarding against bias from a single data split and providing a more accurate reflection of real-world performance.

5.7 Impact of hyperparameter tuning and cross-validation on ML model performance

This section assesses the influence of parameter tuning and cross-validation on the ML models analyzed in this study. These methodologies bolster model precision, generalizability, and resilience by meticulously adjusting parameters and conducting reliable assessments across various data subsets. They are instrumental in mitigating overfitting and in enhancing the consistency and generalization of the models (Li et al., 2023). Table 4 illustrates the R^2 scores for each iteration within the k-fold cross-validation framework, utilizing the training dataset. These scores reflect the models' performance throughout the training phase and encapsulate their ability to explain the variance within the dependent variable.

Table 4 provides key revelations regarding the performance and stability of the models. XGBoost, GBR, and ANN emerged as the top performers, with high mean R^2 values of 0.9289, 0.9290, and 0.8960, respectively, and relatively low standard deviations, indicating stability across the folds. GBR showed exceptional consistency with a standard deviation of 0.0128, while XGBoost demonstrated strong performance with minimal variation (SD = 0.0135). Such consistency across folds implies that these models are not excessively sensitive to the nuances of specific data partitions, thereby enhancing their credibility in extrapolating to new, unobserved datasets.

Conversely, models like Lasso and SVR exhibited lower mean R^2 scores of 0.6144 and 0.8213, with higher variability. Lasso, in particular, had a pronounced standard deviation of 0.0346, signifying a heightened degree of instability in its performance. Despite their diminished accuracy, these models may still possess practical utility or could be enhanced through further optimization. For instance, Lasso could be revisited with alternative regularization strategies, and SVR might see improvements with hyperparameter fine-tuning.

The RF model demonstrated a moderate performance, with a mean R^2 of 0.8637 but a higher SD (0.0223) compared to XGBoost and GBR. This indicates that, while the RF model performs adequately, it is more susceptible to variations in data across folds. KNN, with a mean R^2 of 0.7524, displayed stable but suboptimal performance (SD = 0.0300). The higher standard deviation in KNN indicates that its performance is more influenced by the

Table 4 R^2 Scores for each K-fold for each ML model on training dataset

Model	F 1	F2	F3	F4	F5	Mean	SD
XGBoost	0.9318	0.9398	0.9377	0.9026	0.9327	0.9289	0.0135
GBR	0.9320	0.9430	0.9360	0.9051	0.9292	0.9290	0.0128
RF	0.8876	0.8320	0.8510	0.8621	0.8856	0.8637	0.0223
SVR	0.8555	0.8534	0.8164	0.7707	0.8104	0.8213	0.0313
MLP	0.8832	0.8931	0.8972	0.8496	0.8990	0.8844	0.0182
ANN	0.8964	0.9088	0.8832	0.8848	0.9069	0.8960	0.0107
Lasso	0.6447	0.5730	0.5712	0.6428	0.6403	0.6144	0.0346
KNN	0.7586	0.7024	0.7484	0.7561	0.7963	0.7524	0.0300

specific data split, which may be due to the model's reliance on local patterns in the data.

These results highlight the importance of tuning and cross-validation in minimizing overfitting and improving the models' generalization ability. The fluctuations in R^2 values across the folds reflect the natural variability in cross-validation, where different subsets of the data can lead to slight variations in performance. Such variability is not necessarily a sign of inadequate convergence but rather an intrinsic aspect of the cross-validation methodology. By scrutinizing both the average R^2 and standard deviation, we can attain a more nuanced comprehension of the models' stability and overall predictive prowess. This analysis offers a transparent comparison of the models' capabilities and shortcomings, underscoring the imperative for further refinement and optimization of certain models.

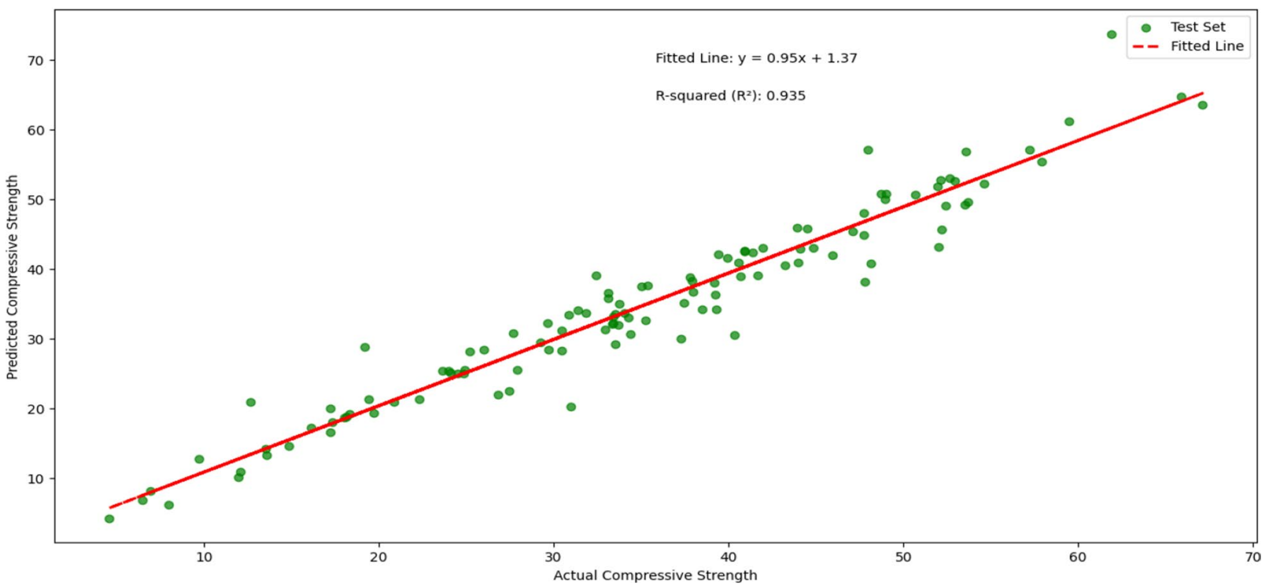

Fig. 11 Alignment between actual and predicted compressive strength in XGBoost model

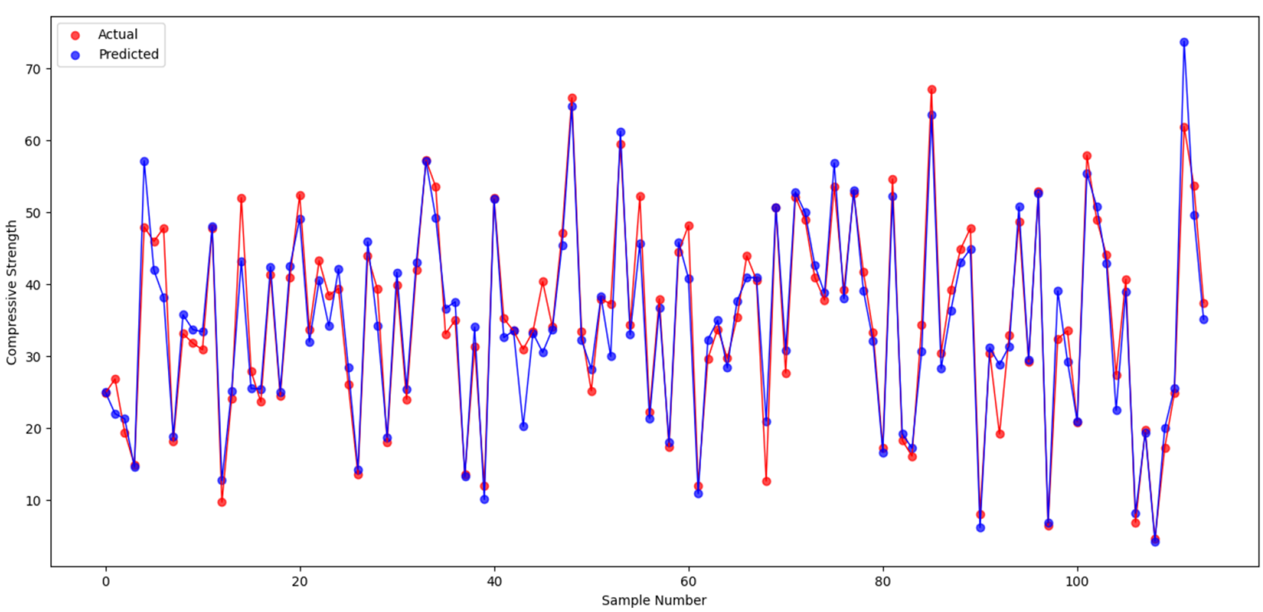

Fig. 12 Scattered plot comparing actual and predicted compressive strength in XGBoost model

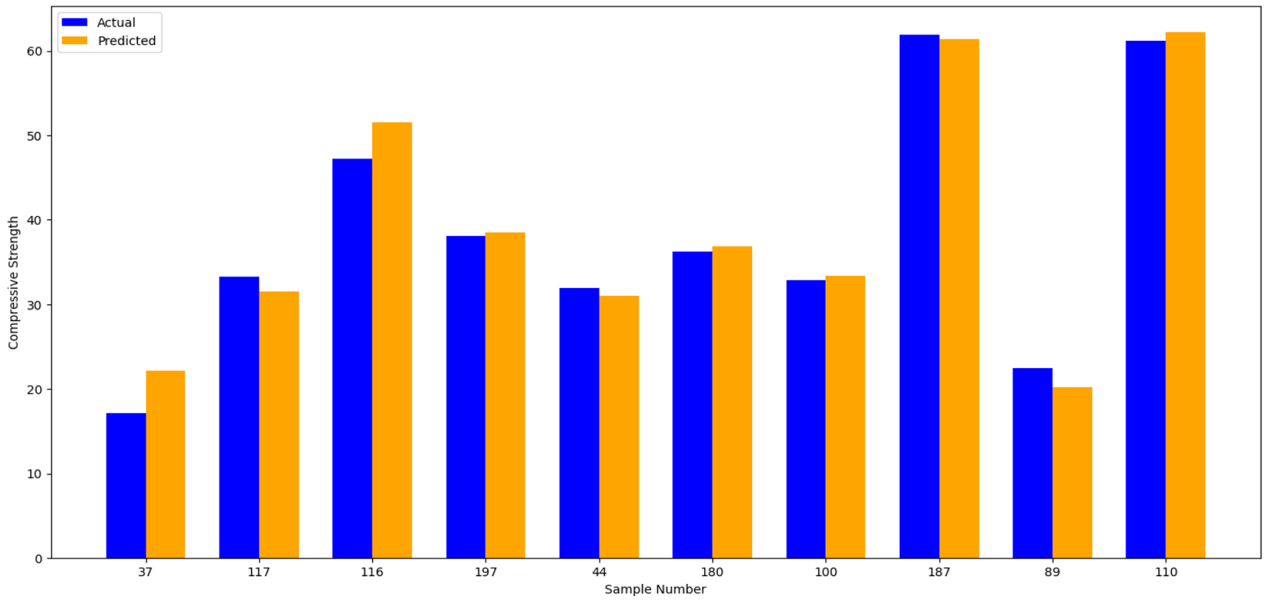

Fig. 13 Juxtaposition bar chart of actual and predicted samples using the XGBoost

5.8 Performance analysis of ML models

This section provides a comprehensive evaluation of the ML models utilized for the prediction of HPC compressive strength. The graphical depictions presented herein meticulously align actual and forecasted values, thereby underscoring the accuracy and dependability of each model. The fluctuation in sample sizes across different algorithms can be attributed to the stochastic nature of the test dataset and the initial conditions during the data partitioning process. This methodology ensures that each model undergoes assessment on a uniform yet unique subset of the data, thereby accentuating their predictive prowess. The analysis underscores the models' proficiency in capturing trends and delivering consistent forecasts, irrespective of the test samples employed.

Figure 11 elucidates the XGBoost model's precision in predicting concrete compressive strength by highlighting the congruence between anticipated and actual values.

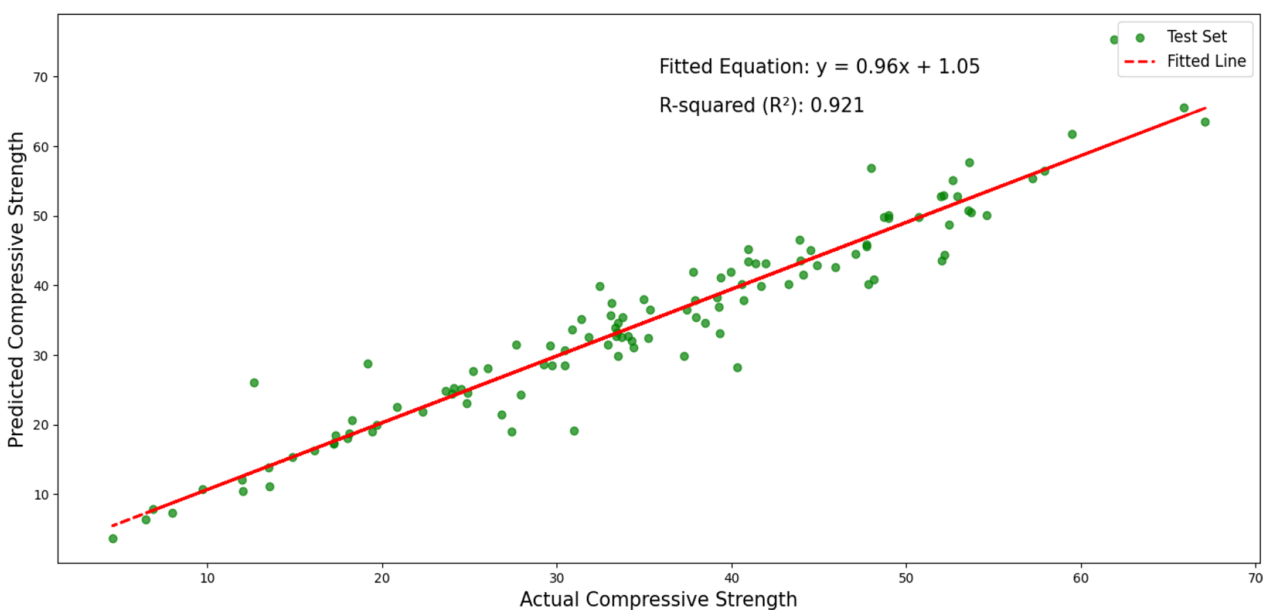

Fig. 14 Alignment between predicted and actual compressive strength in GBR model

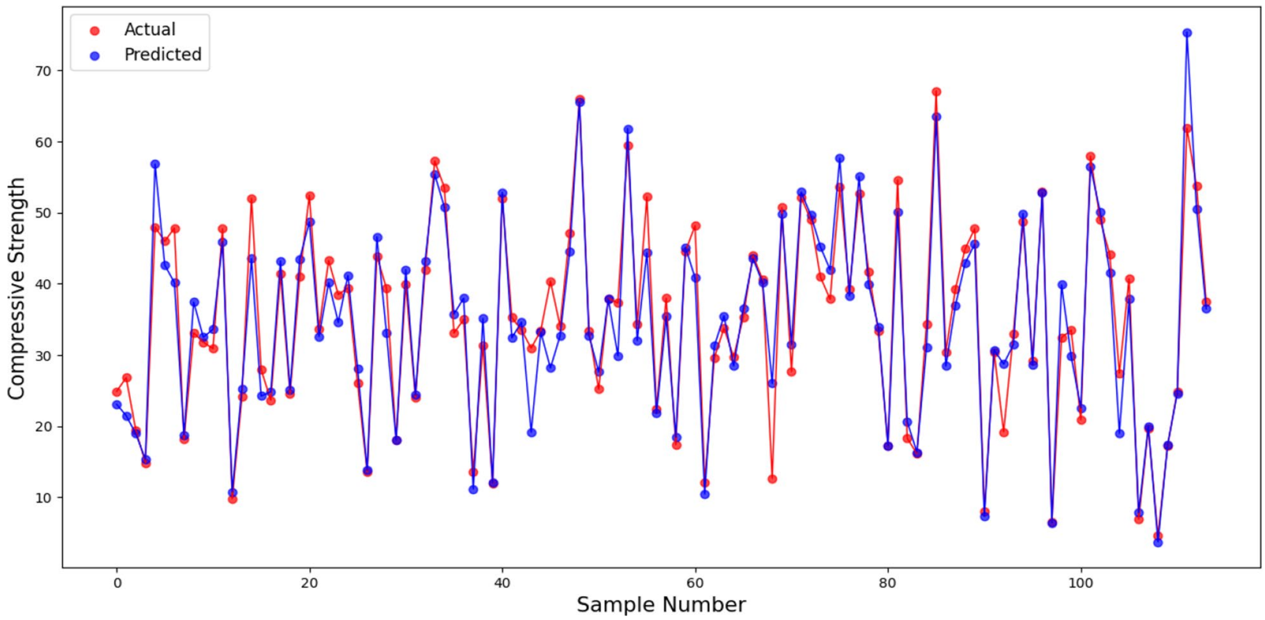

Fig. 15 Scattered plot comparing actual and predicted compressive strength in GBR model

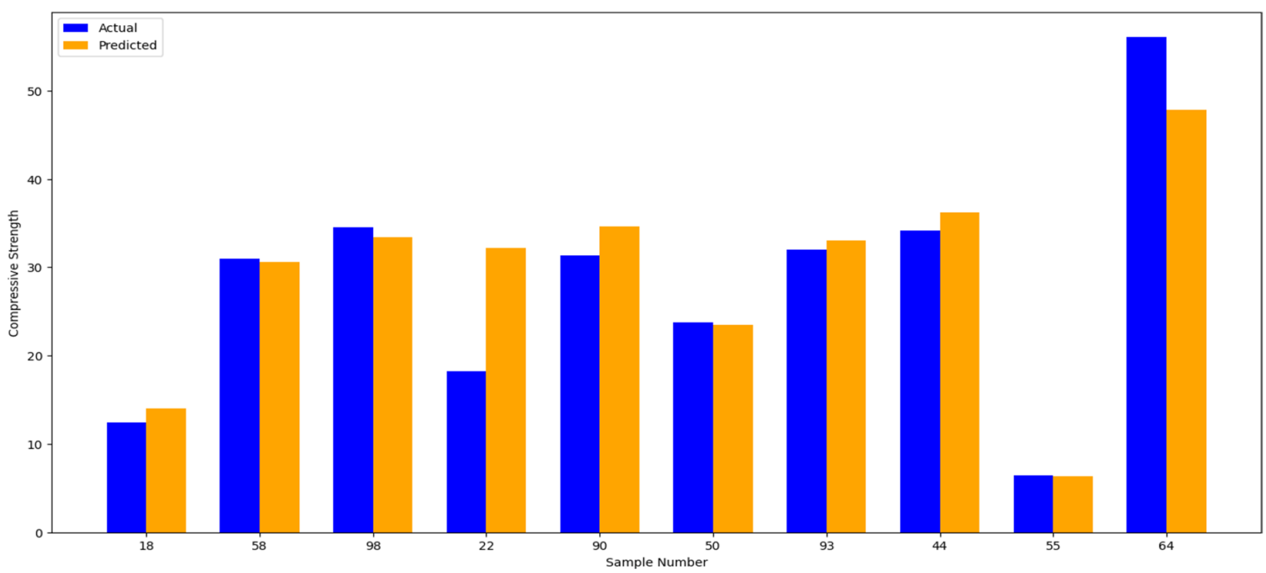

Fig. 16 Juxtaposition bar chart of actual and predicted samples using the GBR

Each emerald dot represents a data point, with the horizontal axis denoting the authentic values from the test sets and the vertical axis reflecting the corresponding forecasts. The alignment of these points delineates a predominantly linear pattern. Superimposed red dashed lines, such as $y = 0.95x + 1.37$ for the test set, which closely align with the ideal $y = x$ relationship, showing minimal deviation. This indicates the model's precision in estimating compressive strength. Figures 12 and 13 expand upon this analysis, delving deeper into the scatter plot patterns and comparing the alignment between actual and predicted values. Collectively, these results confirm XGBoost's effectiveness and reliability in accurately forecasting concrete compressive strength.

Much like the XGBoost model, the GBR algorithm showcased a robust correlation between the values, as illustrated in Fig. 14. The test set fit, represented by green dots, closely aligns with the fitted line, illustrating

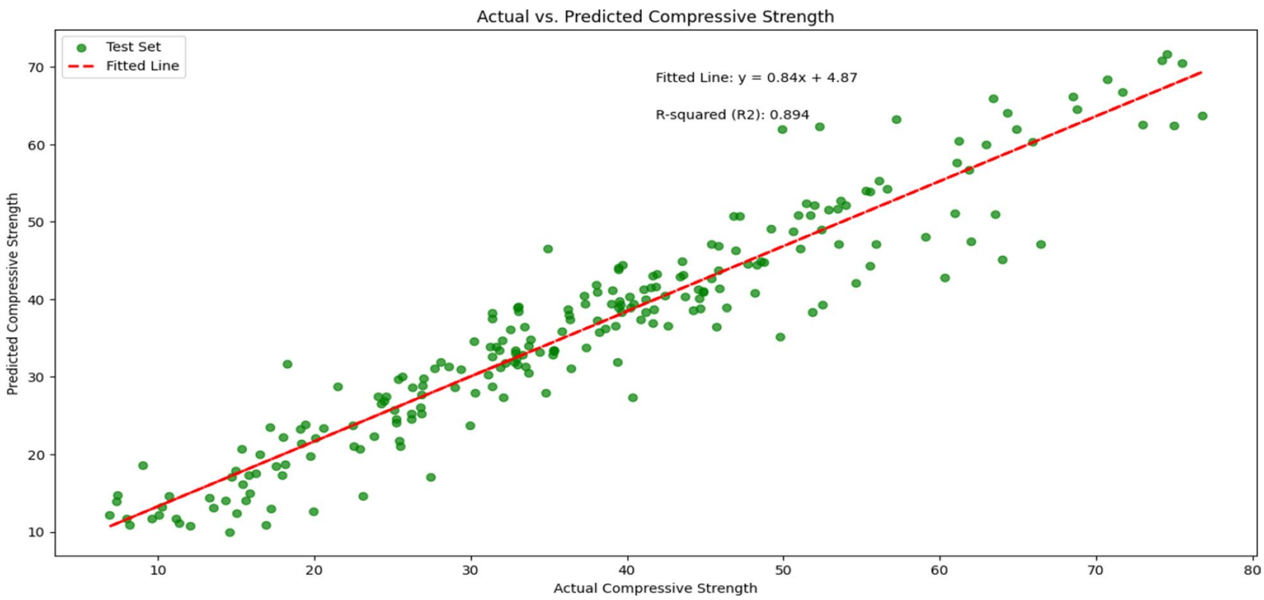

Fig. 17 Alignment between predicted and actual compressive strength in RF model

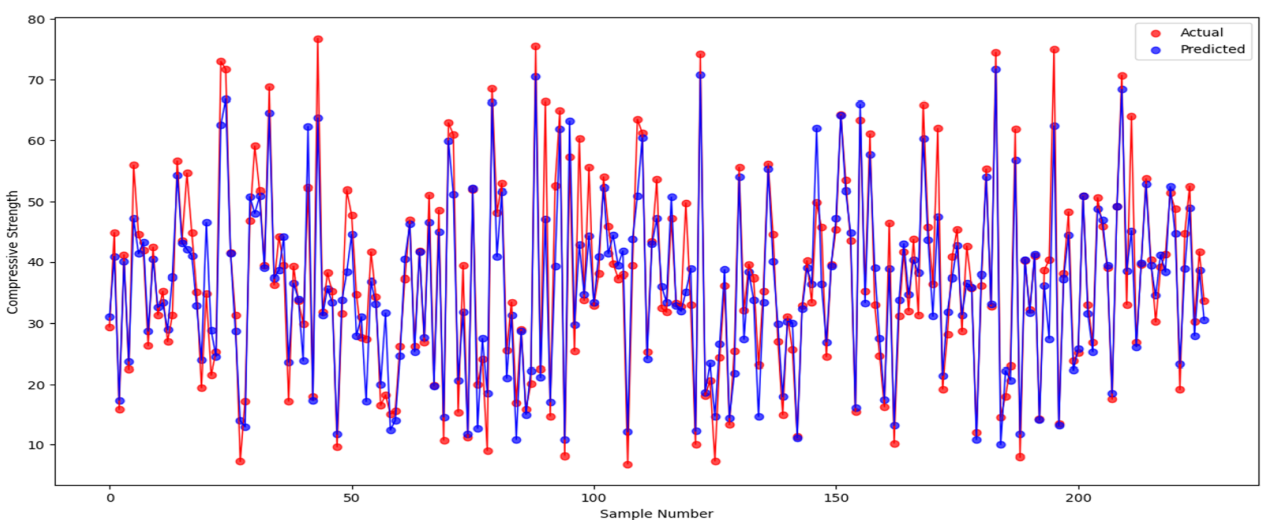

Fig. 18 Scattered plot comparing actual and predicted compressive strength in RF model

the model's capability to understand the connection between predicted and observed values effectively. The fitted equations, such as $y = 0.89x + 4.09$ for the test set, further support this observation. These findings emphasize the GBR model's accuracy in forecasting concrete compressive strength, as supported by the scatter plot portrayed in Figs. 15 and 16, which demonstrates the close alignment between predicted and actual laboratory in the test set. Overall, these results reaffirm the GBR 's precision.

RF model showed a strong correlation between the predicted and actual values, as seen in Fig. 17. The test set aligns closely with the fitted equation $y = 0.84x + 4.87$, indicating reliable predictions. Further analysis, as presented in Figs. 18 and 19, corroborates this consistency. The scatter plots reveal that the red and blue data points are tightly clustered, indicating a minimal divergence from the expected outcomes. Moreover, the bar charts underscore the proximity of actual and predicted values, further validating the model's accuracy. Collectively, the ensemble methods, encompassing RF, XGBoost, and

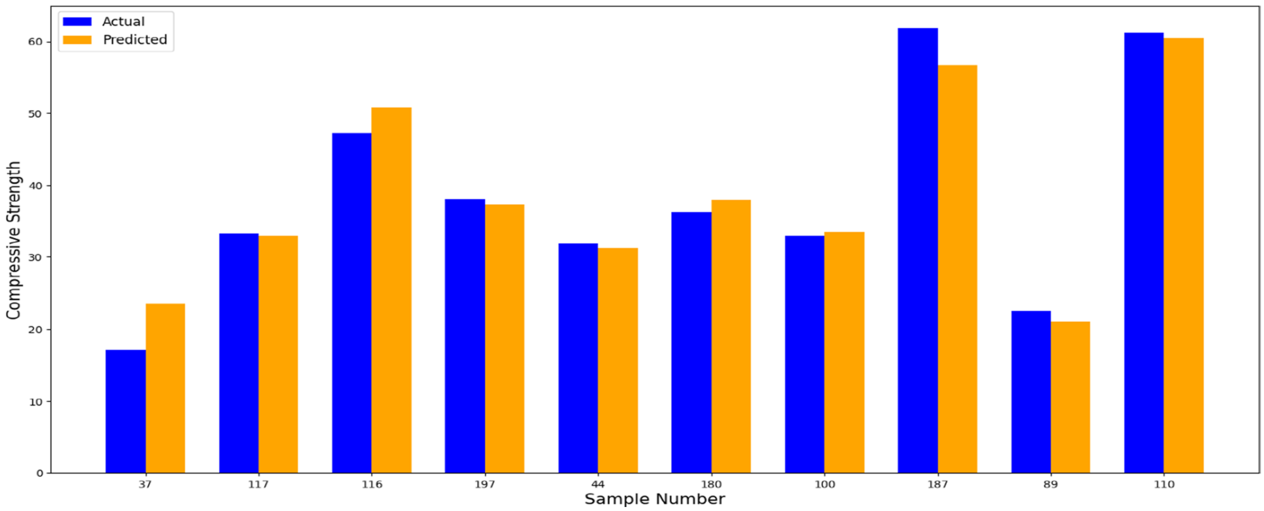

Fig. 19 Juxtaposition bar chart of actual and predicted samples using the RF

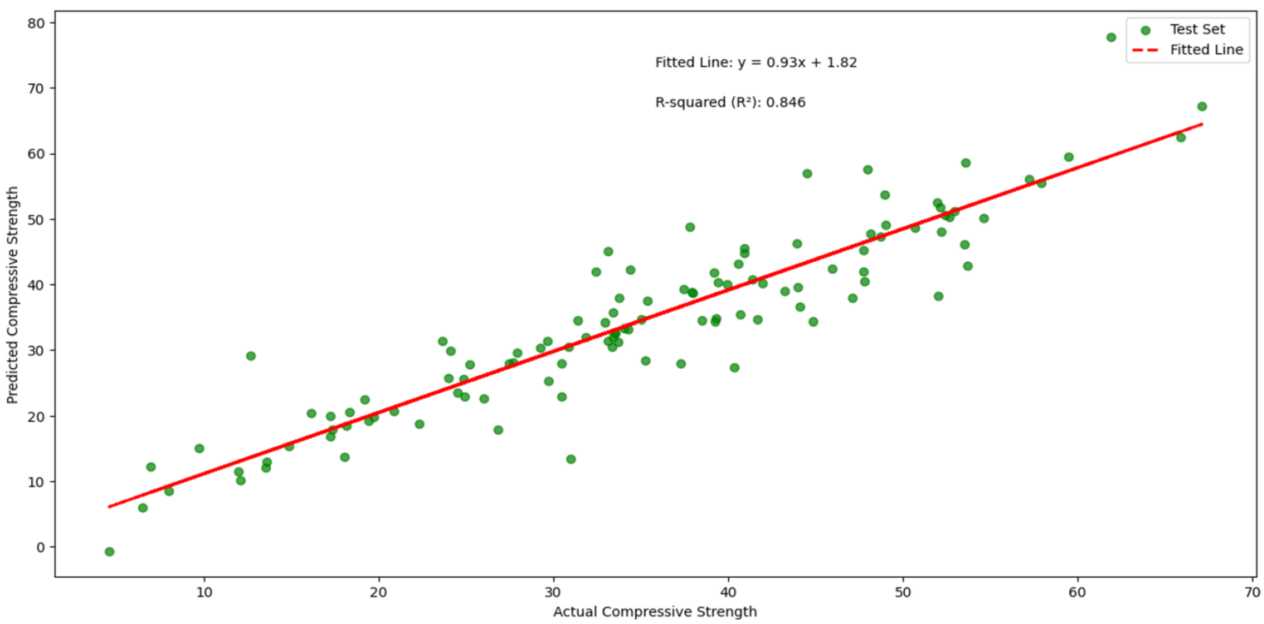

Fig. 20 Alignment between predicted and actual compressive strength in ANN model

GBR, have exhibited outstanding performance, underscoring their efficacy in forecasting.

The ANN algorithm displayed solid performance, with a fitted equation of $y = 0.93x + 1.82$, showcasing a high degree of correlation, as depicted in Fig. 20. The scattered plot in Fig. 21 indicates that the predicted values are closely aligned with the actual values, suggesting minimal error. As a neural network approach, it proved adept at recognizing complex patterns within the data and producing accurate forecasts. However, it exhibited slightly more variability compared to the ensemble models, which could be attributed to the inherent flexibility of neural networks, as illustrated in Fig. 22.

As illustrated in Fig. 23, the MLP model, characterized by the equation $y = 0.87x + 3.68$, exhibited robust predictive capabilities, albeit with a marginally lower accuracy compared to the ANN model. This model demonstrated a slight deviation in the alignment of predicted and observed values, as evidenced by the scatter plot in Fig. 24. The plot indicates that the predicted values are slightly more distant from the actual values than those produced by ensemble methods, hinting at a modest

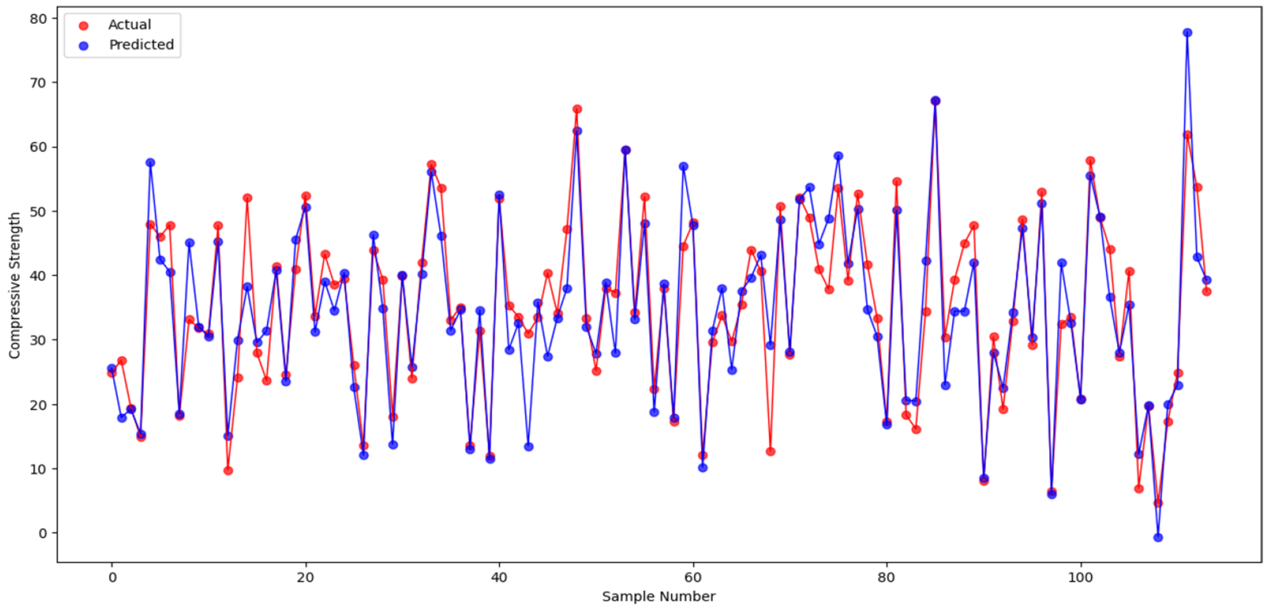

Fig. 21 Scattered plot comparing actual and predicted compressive strength in ANN model

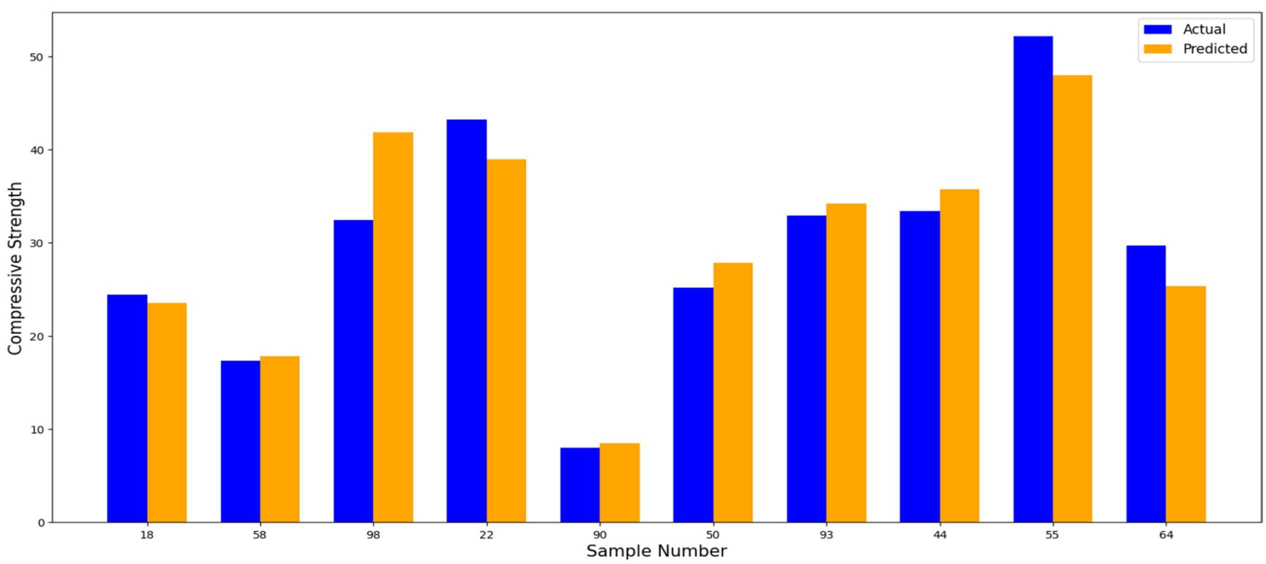

Fig. 22 Juxtaposition bar chart of actual and predicted samples using the ANN

increase in error. Nonetheless, the MLP model maintained a reliable performance, underscoring the effectiveness of neural networks with multiple layers. While both ANN and MLP did not match the precision of the ensemble methods, their strong predictive capabilities highlight their robustness in handling complex data patterns. The bar charts in Fig. 25 further emphasize the minor discrepancy, reaffirming the MLP's reliable performance in forecasting concrete compressive strength. Overall, these results showcase that neural networks, particularly deep architectures, can effectively model concrete strength, even if slightly less precise than ensemble models like XGBoost and GBR.

The efficiency of the SVR, KNN, and Lasso models was notably inferior when compared to ensemble methods, such as XGBoost and GBR. Although these models demonstrated some predictive power, their accuracy was significantly limited, with R^2 values and overall alignment falling short of expectations. In the SVR model, with a fitted equation of ($y = 0.77x + 7.08$), the green dots

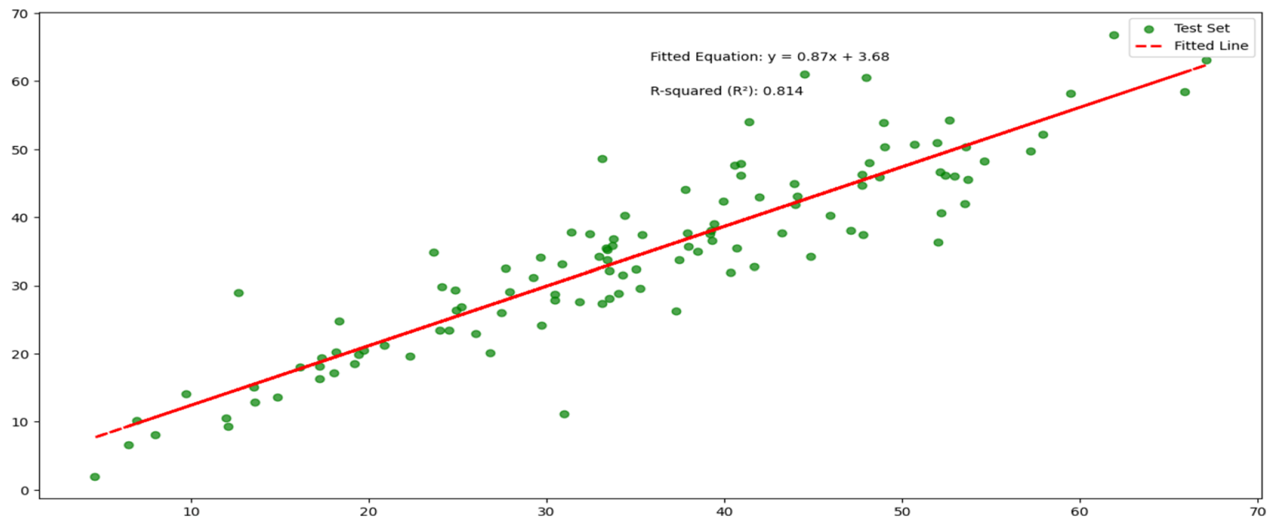

Fig. 23 Alignment between predicted and actual compressive strength in MLP model

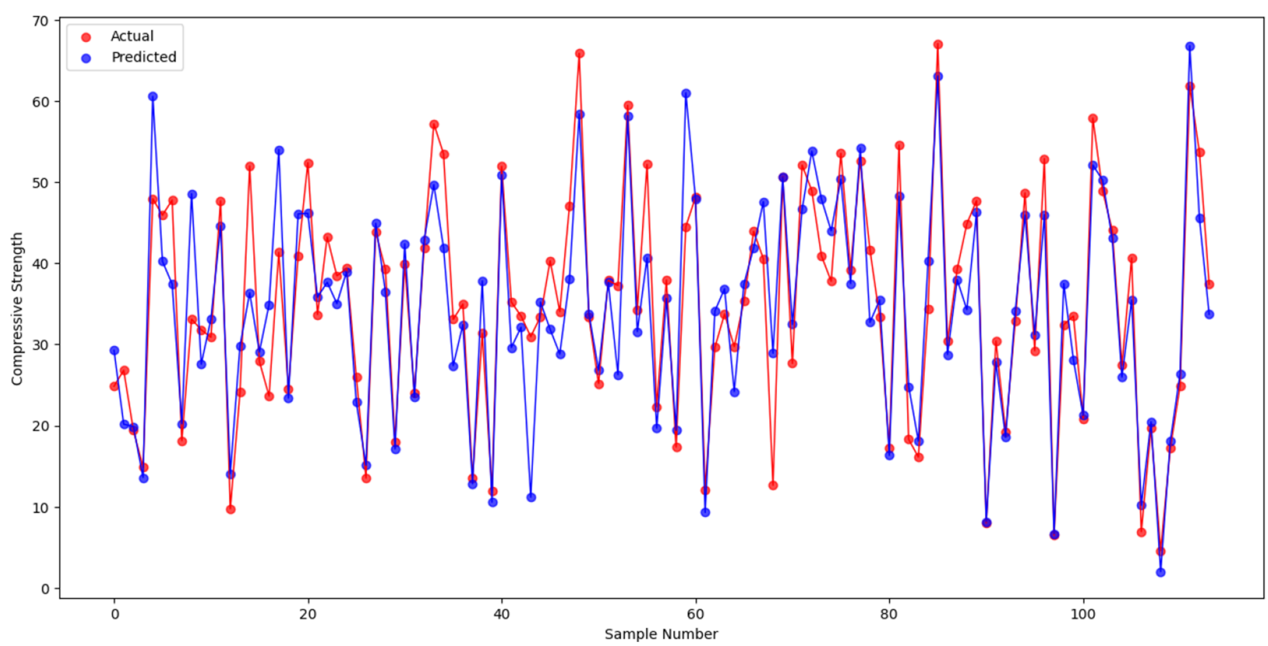

Fig. 24 Scattered plot comparing actual and predicted compressive strength in MLP model

are dispersed around the fitted line, as shown in Fig. 26, thereby highlighting a weaker alignment with the test set. The scattered plot in Fig. 27 illustrates a considerable gap between the observed and forecasted values, reflecting the model's moderate effectiveness in estimating concrete compressive strength. The bar chart in Fig. 28 further highlights this discrepancy.

Similarly, the KNN model, with the equation ($y = 0.68x + 9.51$), faced similar challenges in fitting the data. The alignment shown in Fig. 29 was noticeably less precise compared to the ensemble methods. The scatter plot in Fig. 30 and the bar chart in Fig. 31 further highlight the limited effectiveness of KNN, showing a wider spread between values. This wider discrepancy indicates that KNN had difficulty identifying the complex patterns within the data, resulting in less dependable predictions of compressive strength.

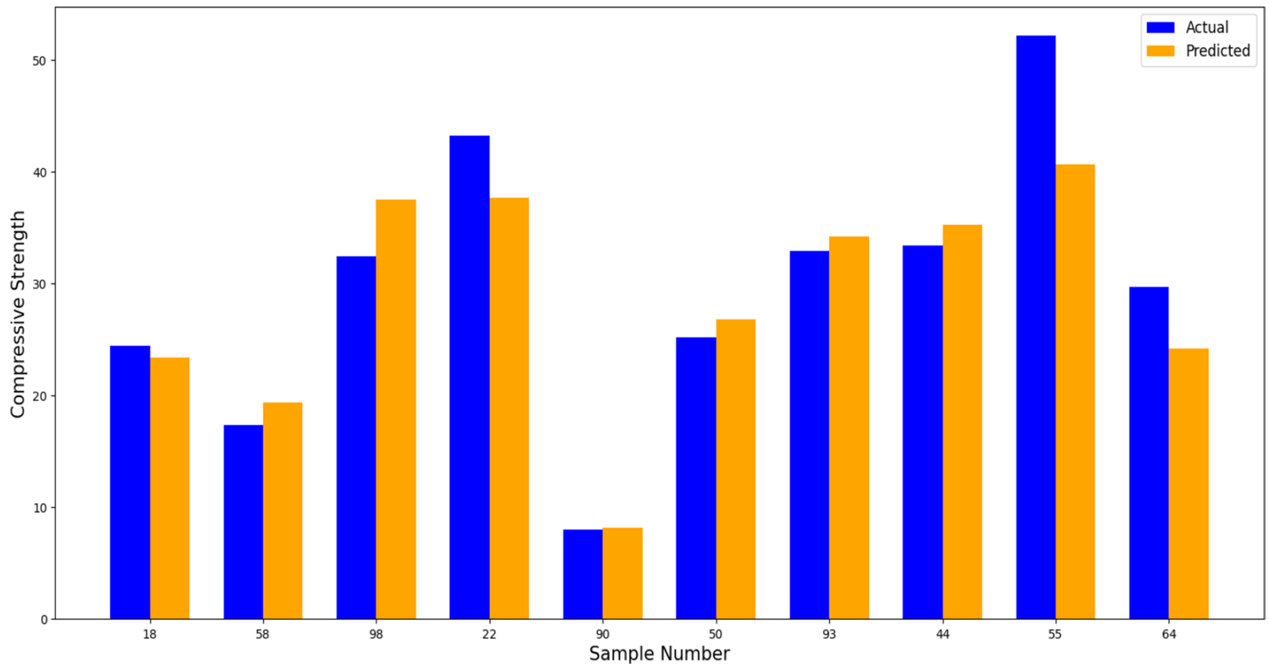

Fig. 25 Juxtaposition bar chart of actual and predicted samples using the MLP

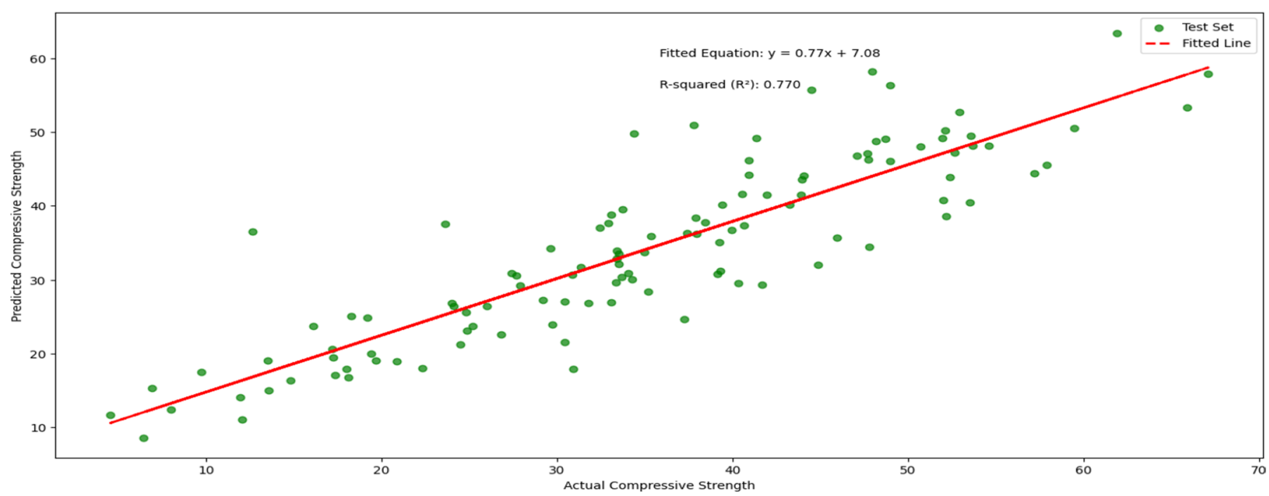

Fig. 26 Alignment between predicted and actual compressive strength in SVR model

Lasso, however, was the least effective among the eight algorithms, with a fitted equation of ($y = 0.58x + 14.90$). As illustrated in Fig. 32, the fit was particularly poor, with the data points deviating significantly from the regression line. The scatter plot in Fig. 33 and the comparison chart in Fig. 34 reveal a considerable gap, highlighting the model's inability to capture underlying trends in the data. The bar chart further accentuates this issue, displaying considerable differences in strength values. These findings emphasize that Lasso not only lagged behind the ensemble models but also performed inferiorly to both SVR and KNN, positioning it as the least reliable model for forecasting compressive strength.

6 Limitations

Despite the promising results and insights gained from this study, certain limitations were identified that could influence the overall findings and their applicability. These limitations primarily stem from dataset

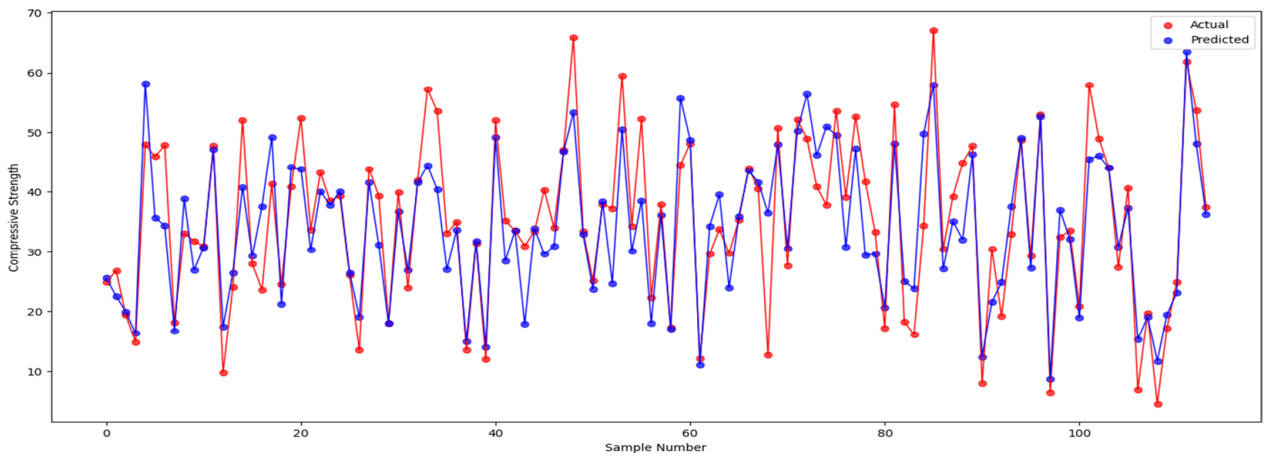

Fig. 27 Scattered plot comparing actual and predicted compressive strength in SVR model

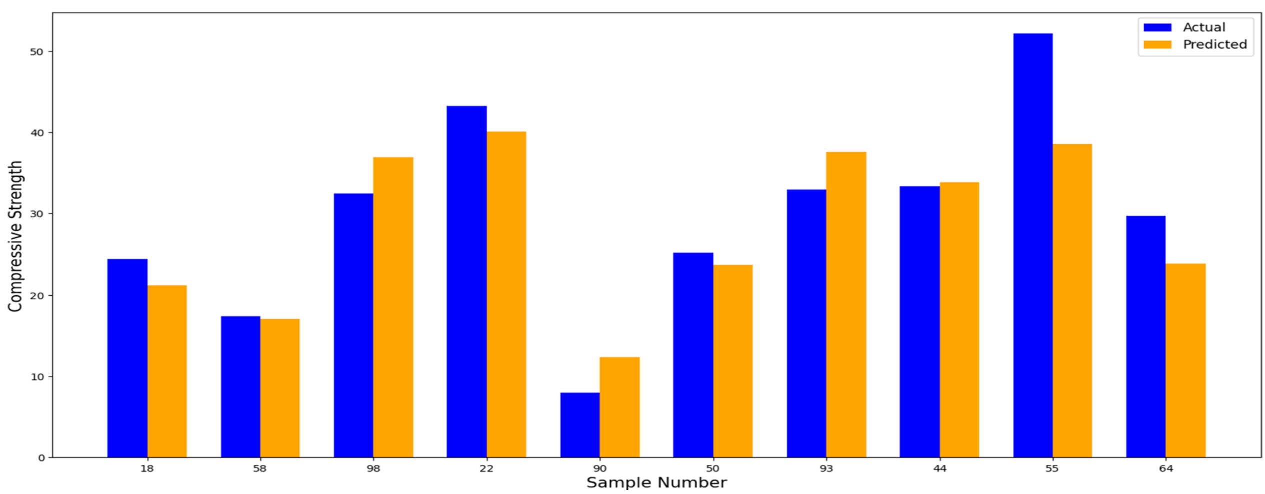

Fig. 28 Juxtaposition bar chart of actual and predicted samples using the SVR

constraints, computational challenges, and the inherent complexity of modeling concrete strength, as outlined below.

1. Although the dataset used in this study was detailed, it neglected critical factors impacting concrete strength, such as detailed curing conditions and environmental factors. Including these variables would enrich the analysis and further improve the predictive accuracy of the models.
2. The influence of the heat of hydration, a critical factor in concrete performance, was not accounted for in the dataset. Variations in heat generation during curing can significantly affect strength development, and its exclusion may impact the accuracy of the predictive models.
3. Hyperparameter tuning for complex models proved to be computationally expensive, requiring significant time resources. This limitation highlights the need for more efficient tuning methods or the integration of an automated machine learning (AutoML) pipeline to streamline the process.

7 Conclusions and future work

This research provides a comprehensive assessment of eight ML algorithms for predicting HPC compressive strength, with particular relevance to construction industry applications. Through systematic evaluation of

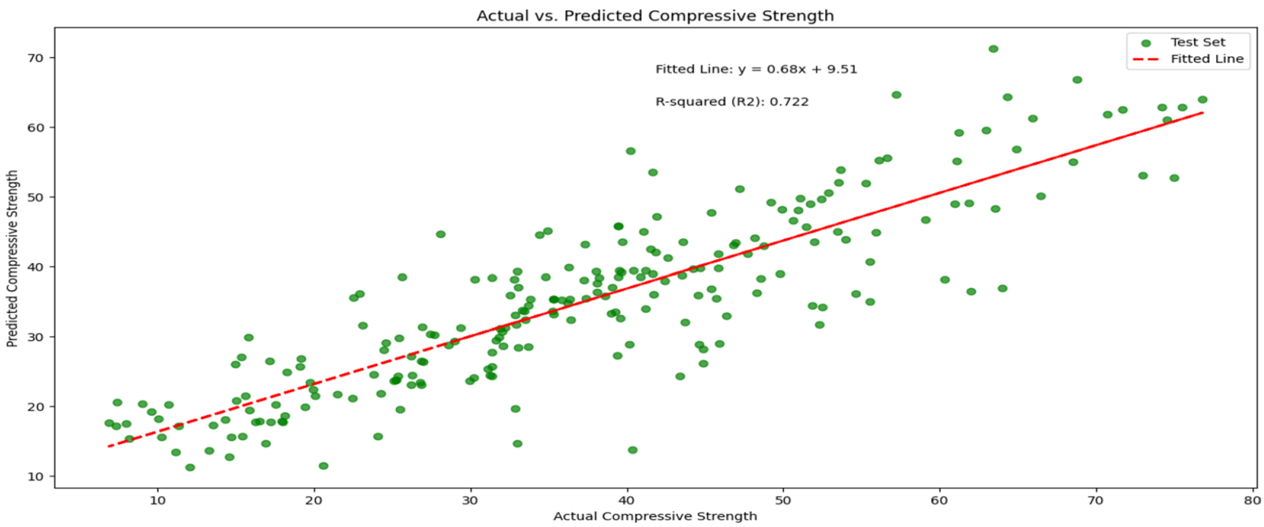

Fig. 29 Alignment between actual and predicted values from the KNN model

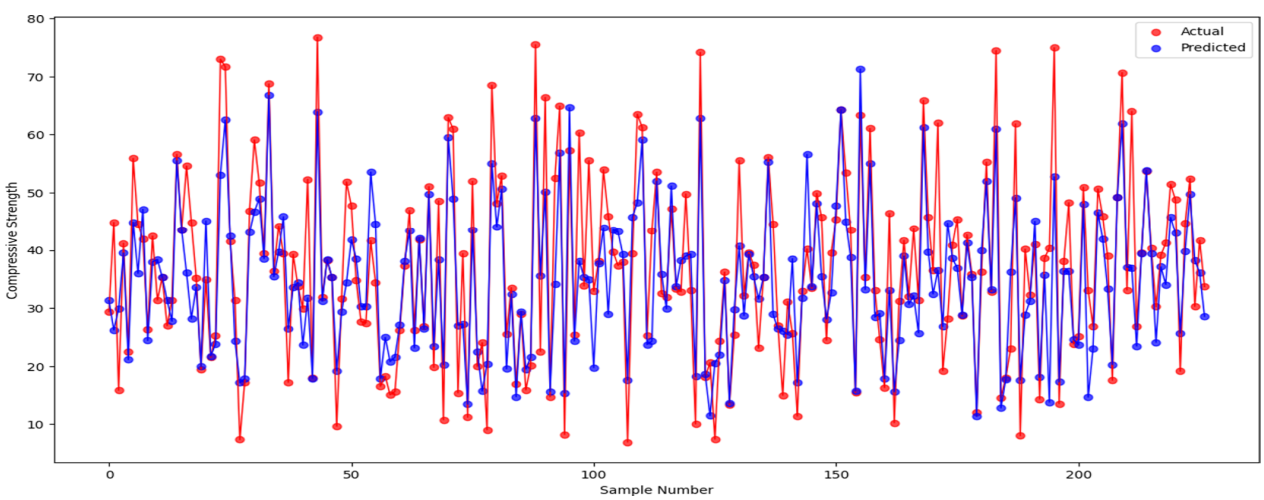

Fig. 30 Scattered plot between actual and predicted values from the KNN model

XGBoost, GBR, RF, SVR, KNN, Lasso, ANN, and MLP models, the study delineates their respective strengths and limitations in concrete strength forecasting. SHAP analysis further elucidates input feature impacts, yielding actionable insights for mix design optimization. The principal findings and implications are as follows:

1. *Model Performance Hierarchy* XGBoost and GBR demonstrated superior efficacy, achieving test accuracies of 93.49% and 92.09% respectively, while RF, ANN, and MLP showed competent performance. These ensemble and neural network approaches substantially outperformed SVR, KNN, and Lasso, with Lasso exhibiting the lowest accuracy (56.11%). The success of XGBoost and GBR stems from their ability to capture complex data interactions.
2. *Computational Efficiency* XGBoost balanced accuracy with computational efficiency, completing training in 0.12 s. Although ANN and MLP required longer training durations, KNN and Lasso prioritized speed over precision, underscoring XGBoost's optimal performance profile.
3. *Generalization Capacity* On training data, XGBoost ($R^2 = 0.9289$) and GBR ($R^2 = 0.9290$) maintained strong predictive fidelity with minimal overfitting, contrasting with SVR, KNN, and Lasso's elevated

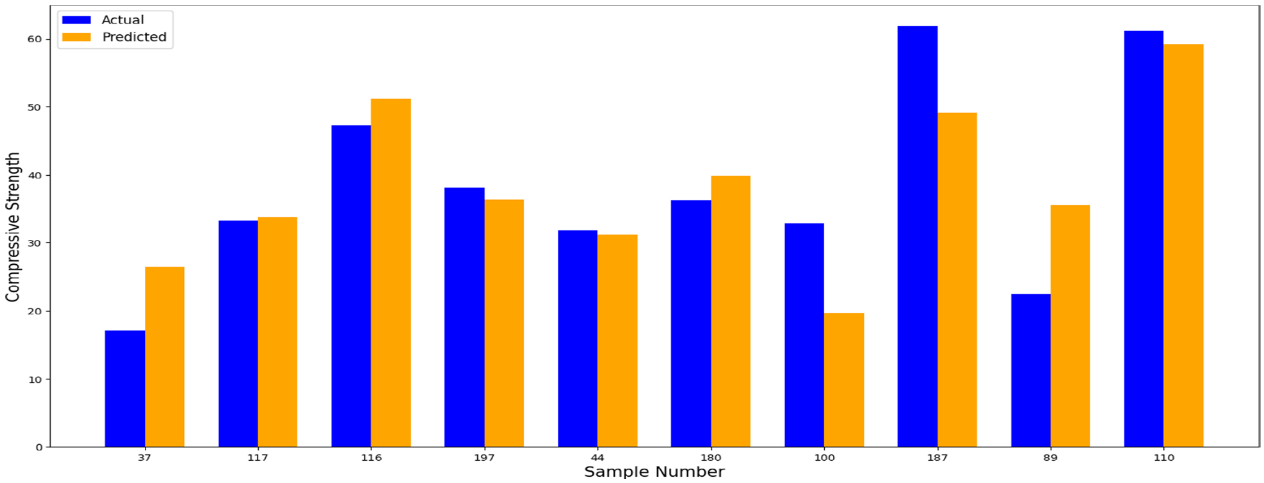

Fig. 31 Juxtaposition bar chart of actual and predicted samples using the KNN

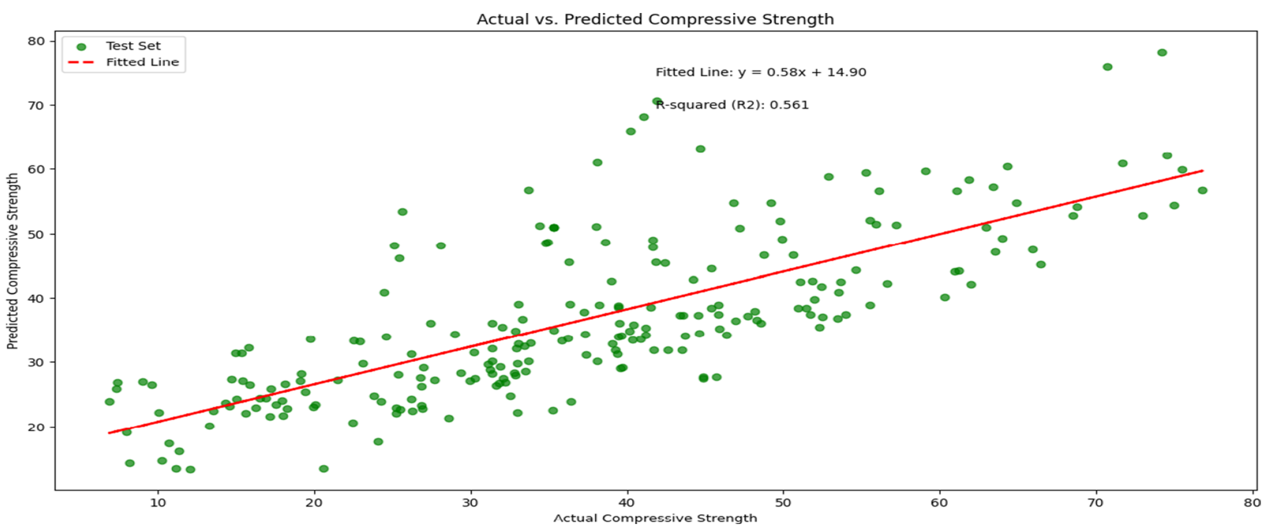

Fig. 32 Alignment between actual and predicted values from the Lasso model

error margins. This reaffirms ensemble methods' reliability for concrete strength prediction.

4. *Optimization Impact* Hyperparameter tuning substantially enhanced model performance, particularly for XGBoost and GBR. Cross-validation effectively mitigated overfitting/underfitting risks, ensuring robust generalization assessments.
5. *Parameter Significance* Correlation analyses (Pearson/Kendall Tau) identified age and cement content as critical determinants of compressive strength. Pearson coefficients revealed strong linear relationships, while Kendall Tau validated these through nonlinear interaction analysis.
6. *Feature Interpretation* SHAP analysis confirmed age (mean SHAP + 8.18) and cement (+ 6.92) as dominant predictors, with supplementary materials (slag, superplasticizers) exhibiting moderate influence. These findings enable targeted mix design optimization.

This study offers valuable guidance for selecting ML models to predict HPC compressive strength, providing insights into model performance, optimization strategies, and potential challenges such as parameter tuning, computational costs, and data quality. The demonstrated framework enhances predictive accuracy while

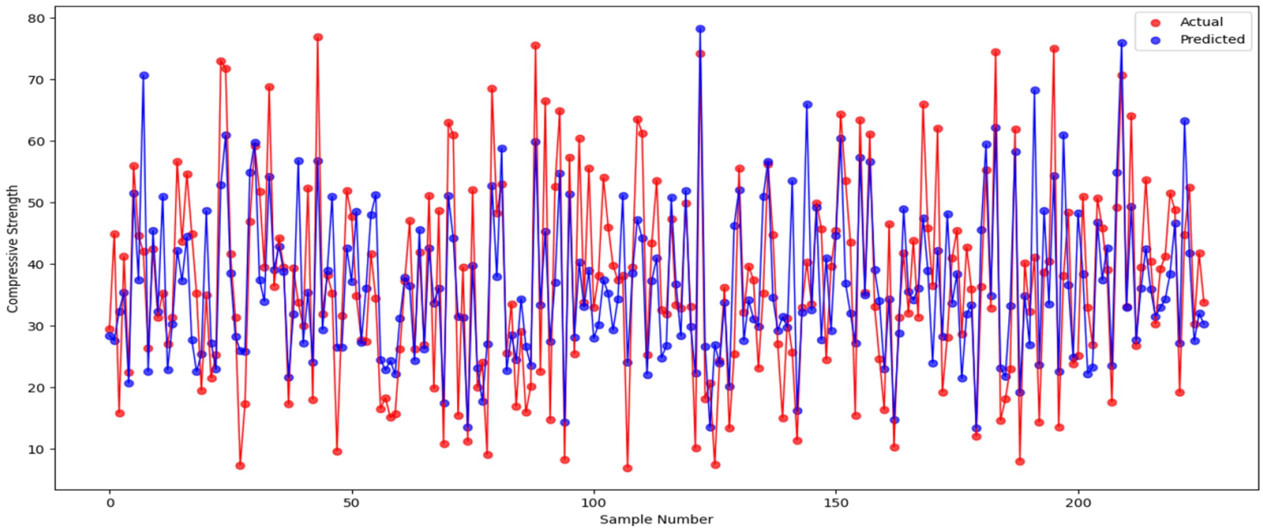

Fig. 33 Comparison between actual and predicted values from the Lasso model

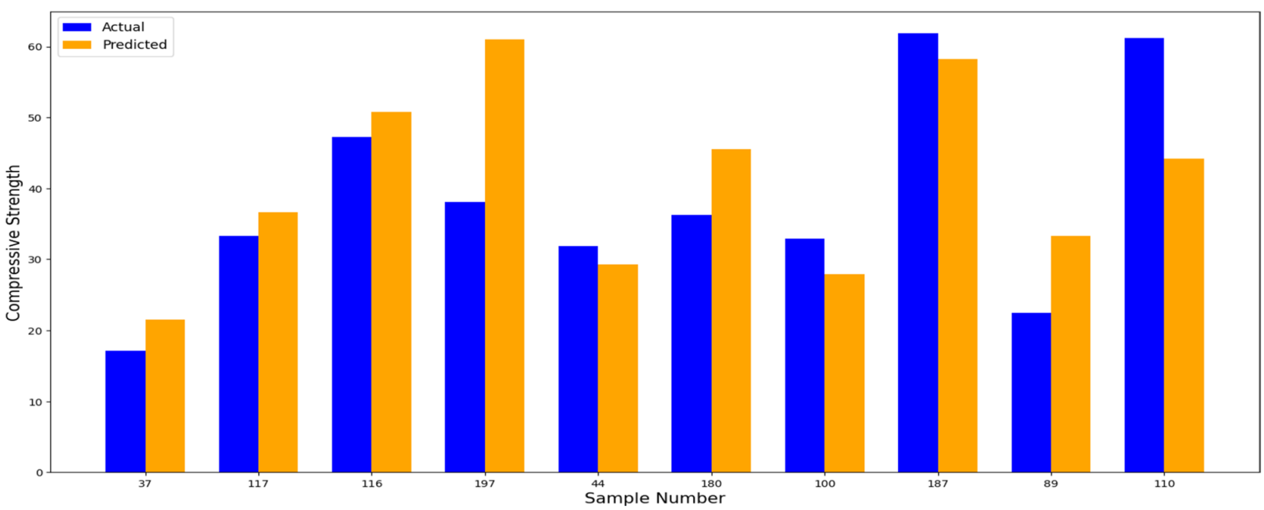

Fig. 34 Juxtaposition bar chart of actual and predicted samples using the Lasso

addressing operational challenges in computational cost and experimental design.

Looking ahead, enhancing prediction accuracy in concrete strength modeling is crucial for advancing construction practices. Future research should explore advanced model architectures and domain-specific features to achieve this goal. Experimenting with other ensemble methods, such as AdaBoost, CatBoost, or LightGBM, along with deep learning techniques like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), could significantly boost prediction accuracy by leveraging diverse data representations and complex pattern recognition capabilities. Additionally, incorporating domain-specific features, such as detailed concrete composition, curing conditions, environmental influences, and construction practices, can enrich the input data and enhance model predictions. To handle larger datasets and facilitate real-time applications, advancements in distributed computing frameworks like Apache Spark and parallel processing optimizations are essential. These improvements can enhance computational efficiency and enable the analysis of extensive datasets, which is vital for the construction industry.

Also, AutoML pipelines, such as TPOT and Auto-Sklearn, offer promising avenues for automating the ML process, significantly reducing the time and effort required to build models. These pipelines could be applied to concrete strength prediction to streamline hyperparameter tuning and feature engineering. However, AutoML technologies face challenges, including handling complex data, ensuring model interpretability, and managing the risk of overfitting. Although not yet fully stable, they represent a promising area for future research as the technology matures.

Furthermore, adopting advanced thermal sensing technologies, such as Fiber Bragg Grating (FBG) sensors, to monitor heat of hydration across various mix designs and sample sizes could greatly enhance real-time strength prediction. Given that hydration heat is a crucial factor influencing concrete strength, accurately capturing its variations can provide deeper insights into strength development. Integrating these sensors with predictive models would refine accuracy, optimize curing conditions, and improve data-driven decision-making in concrete mix design. Synergistic combination of sensor networks with predictive models promises more accurate, efficient, and scalable solutions for next-generation construction practices.

Acknowledgements
The authors sincerely thank the University of Ilorin, the Faculty of Engineering, and the Department of Civil Engineering, along with the Department Chair, for their unwavering support and resources throughout this research.

Institutional review board statement
Not applicable.

Author contributions
Methodology, S.O.A.; validation, S.O.A.; formal analysis, S.O.A.; investigation, S.O.A., A.N.S., O.A.A. and B.J.B.; resources, Y.O.B.; data curation, Y.O.B., S.O.A., A.N.S., O.A.A., and B.J.B.; writing— original draft preparation, S.O.A., A.N.S., O.A.A., and B.J.B.; writing—review and editing, S.O.A. and O.A.A.; visualization, S.O.A.; supervision, Y.O.B. All contributors have reviewed and approved the final version of the manuscript for publication.

Funding
This research received no external funding.

Data availability
The dataset (YEH), which includes all the features referenced in this study, is accessible in the public UCI Irvine ML Repository at the following link: YEH.

Declarations

Informed consent
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 11 December 2024 Revised: 8 March 2025 Accepted: 24 April 2025
Published online: 01 July 2025

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